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DPmixGPD: Dirichlet Process Mixtures with Generalized Pareto Tails in R

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Abstract

We present DPmixGPD, an R package for Bayesian nonparametric mixture modeling with explicit extreme-value tail behavior. The package models the bulk distribution using a Dirichlet process mixture (DPM) with multiple kernel families and supports both Chinese restaurant process (CRP) and truncated stick-breaking (SB) backends. For heavy-tailed outcomes, DPmixGPD optionally splices a generalized Pareto distribution (GPD) above a threshold, enabling coherent extrapolation for extreme quantiles and tail probabilities while retaining flexible bulk fit. The workflow is designed around two main functions: build-nimble-bundle constructs a compiled NIMBLE model bundle, and run-mcmc-bundle-manual runs MCMC to produce a fitted object with print, summary, plot, fitted, and predict methods. We demonstrate density estimation, conditional modeling with covariates, and a real-data application using the built-in airquality dataset, highlighting stable high-quantile inference and tail exceedance summaries.

Keywords: —!!!—at least one keyword is required—!!!—.

1. Introduction

Real-world data are often shaped by two competing demands: (i) the *bulk* of the distribution can exhibit complex structure such as skewness or multimodality, and (ii) the *tail* may contain extreme outcomes that require principled extrapolation beyond observed values. Bayesian nonparametric mixture models address (i) by allowing flexible densities without fixing the number of components in advance, typically through a Dirichlet process (DP) prior (Ferguson 1973; Neal 2000). Extreme value theory addresses (ii) by characterizing exceedances above a high threshold via the generalized Pareto distribution (GPD) (Pickands 1975; de Haan and Ferreira 2006). The **DPmixGPD** package combines these ideas in a unified Bayesian framework. The bulk distribution is modeled as a Dirichlet process mixture (DPM)

with selectable kernels and backends. Optionally, the upper tail is modeled using a GPD beyond a threshold, yielding a spliced distribution that preserves bulk flexibility while enforcing extreme-value tail behavior. This design targets settings where bulk-only mixtures can be sensitive in extreme extrapolation (e.g., unstable very-high quantiles), and where purely parametric bulk+tail models are too rigid for the central distribution. The package emphasizes a practical workflow:

1. `build_nimble_bundle()` constructs a **NIMBLE**-based model bundle from data and a model specification.

2. `run_mcmc_bundle_manual()` runs MCMC and returns a fitted object with S3 methods for summarization, diagnostics, and prediction.

Users can compute posterior predictive densities, survival curves, and quantiles (including extreme quantiles) with coherent uncertainty quantification.

2. Model formulation

2.1. Bulk model: Dirichlet process mixtures

Let $y_1, \ldots, y_n$ denote observations and $f(\cdot \mid \theta)$ a kernel density (e.g., lognormal, gamma, normal, Laplace, inverse Gaussian, Amoroso). A Dirichlet process mixture model can be written hierarchically as

$$y_i \mid \theta_i \sim f(y_i \mid \theta_i), \tag{1}$$
$$\theta_i \mid G \sim G, \tag{2}$$
$$G \sim \mathrm{DP}(\alpha, G_0), \tag{3}$$

where α is the concentration parameter and G_0 is a base distribution for kernel parameters. Marginally, this implies an (infinite) mixture representation

$$p(y) = \sum_{j=1}^{\infty} w_j f(y \mid \Theta_j),$$

with random weights w_j and atoms Θ_j induced by the DP prior.

2.2. Inference backends: CRP and stick-breaking

DPmixGPD supports two standard representations:

CRP backend. A Chinese restaurant process (CRP) sampler integrates out G and samples cluster allocations. This representation is convenient when the number of occupied clusters is unknown and allows it to adapt automatically (Neal 2000).

SB backend. A truncated stick-breaking (SB) representation approximates the infinite mixture with J components, where weights follow a stick-breaking prior and J is chosen sufficiently large. The posterior typically concentrates on a smaller effective number of components.

2.3. Tail model: generalized Pareto distribution

To model extremes, **DPmixGPD** optionally splices a generalized Pareto distribution above a threshold u. Let $F_{\text{bulk}}(y)$ denote the bulk CDF. For $y > u$, define

$$F(y) = F_{\text{bulk}}(u) + \{1 - F_{\text{bulk}}(u)\} F_{\text{GPD}}(y; u, \sigma, \xi),$$

where $F_{\text{GPD}}(\cdot)$ is the GPD CDF with scale $\sigma > 0$ and shape ξ (Pickands 1975; de Haan and Ferreira 2006). This formulation ensures a valid distribution with continuity at u and interpretable tail behavior: $\xi > 0$ corresponds to heavy tails, $\xi = 0$ to exponential-like tails, and $\xi < 0$ to bounded tails. The threshold u can be fixed or inferred. In many applications, the threshold is treated as unknown with a weakly informative prior centered near a high empirical quantile. The resulting posterior uncertainty in u is then propagated into tail quantiles and exceedance probabilities.

2.4. Conditional modeling with covariates

When covariates x_i are available, **DPmixGPD** supports conditional modeling where selected kernel parameters (and optionally tail components) depend on x_i through link functions. In practice, `build_nimble_bundle()` accepts a design matrix `X` and constructs a conditional mixture model so that predictions can be computed for new covariate profiles.

3. Implementation

DPmixGPD is implemented in R and uses **NIMBLE** to express and compile probabilistic models. The package contains:

- A kernel registry with supported distributions for bulk modeling.
- Custom likelihood components for mixture densities and mixture+GPD splices.
- A model builder that generates **NIMBLE** code from a user specification, including priors and backends.
- An MCMC runner that executes compiled samplers and returns a fitted object with S3 methods.

The API is designed to be stable: the same user-facing functions cover unconditional, conditional, and tail-augmented models with minimal changes in arguments.

4. Basic usage

This section illustrates the typical workflow. We keep the example light enough to render quickly, but in real analysis users should increase iterations and check convergence diagnostics carefully.

4.1. Unconditional fit

```
library("DPmixGPD")

set.seed(1)
y <- abs(rnorm(80)) + 0.2

bundle <- build_nimble_bundle(
  y = y,
  backend = "sb",
  kernel  = "normal",
  GPD     = TRUE,
  components = 8,
  mcmc = list(niter = 200, nburnin = 50, thin = 2, nchains = 1, seed = 1)
)

fit <- run_mcmc_bundle_manual(bundle, show_progress = FALSE)

## ===== Monitors =====
## thin = 1: alpha, mean, sd, tail_scale, tail_shape, threshold, w, z
## ===== Samplers =====
## RW sampler (27)
##   - alpha
##   - mean[]  (8 elements)
##   - sd[]  (8 elements)
##   - threshold
##   - tail_scale
##   - tail_shape
##   - v[]  (7 elements)
## categorical sampler (80)
##   - z[]  (80 elements)

print(fit)

## MixGPD fit | backend: Stick-Breaking Process | kernel: Normal Distribution | GPD tail:
## n = 80 | components = 8 | epsilon = 0.025
## MCMC: niter=200, nburnin=50, thin=2, nchains=1
## Fit
## Use summary() for posterior summaries; plot() for diagnostics; predict() for prediction

summary(fit)

## MixGPD summary | backend: Stick-Breaking Process | kernel: Normal Distribution | GPD ta
## n = 80 | components = 8
## Summary
## Initial components: 8 | Components after truncation: 3
```

```
##
## WAIC: 67.076
## lppd: -8.973 | pWAIC: 24.565
##
## Summary table
##   parameter   mean    sd q0.025 q0.500 q0.975    ess
##  weights[1]  0.467 0.029  0.411  0.462  0.514 41.146
##  weights[2]  0.324 0.059  0.212  0.325  0.438  3.292
##  weights[3]  0.138 0.041  0.062  0.138  0.225 16.443
##       alpha  0.740 0.347  0.225  0.710  1.479 13.539
##  tail_scale  0.603 0.067  0.491  0.568  0.708  5.501
##  tail_shape -0.072 0.098 -0.218 -0.086  0.087  5.597
##   threshold  0.734 0.009  0.721  0.733  0.743  2.880
##     mean[1]  0.942 1.719  0.440  0.455  7.337 17.237
##     mean[2]  5.202 2.676  0.455  5.037  9.817  9.531
##     mean[3]  5.579 2.976  1.209  5.123 10.811 31.211
##       sd[1]  0.317 0.397  0.180  0.222  1.766 21.288
##       sd[2]  1.214 0.649  0.180  1.181  2.703 35.086
##       sd[3]  1.511 0.921  0.142  1.282  3.533 47.265
```

4.2. Posterior prediction

The `predict()` method can return posterior predictive densities, survival functions, and quantiles. Exact return formats depend on the package version; the goal below is to demonstrate typical calls.

```
# Predict selected quantiles
pred_q <- predict(fit, type = "quantile", index = c(0.90, 0.95, 0.99))
pred_q

## $fit
##   estimate index    lower    upper
## 1 1.679593  0.90 1.477805 1.786626
## 2 2.043072  0.95 1.765745 2.245004
## 3 2.835694  0.99 2.367136 3.397640
##
## $lower
## NULL
##
## $upper
## NULL
##
## $type
## [1] "quantile"
##
## $grid
```

```
## [1] 0.90 0.95 0.99
##
## $draws
##          [,1]     [,2]     [,3]     [,4]     [,5]     [,6]     [,7]     [,8]
## [1,] 1.692930 1.681223 1.675952 1.693834 1.689514 1.594770 1.622196 1.794559
## [2,] 2.036403 2.025581 2.020708 2.037238 2.033245 1.873125 1.949708 2.244158
## [3,] 2.737107 2.728090 2.724030 2.737803 2.734476 2.376644 2.627512 3.354461
##          [,9]    [,10]    [,11]    [,12]    [,13]    [,14]    [,15]    [,16]
## [1,] 1.745793 1.743019 1.764997 1.639890 1.621812 1.473205 1.704564 1.727082
## [2,] 2.193556 2.190677 2.213483 2.034004 2.015245 1.729493 2.043349 2.063522
## [3,] 3.299326 3.296189 3.321038 3.007284 2.986844 2.226324 2.700104 2.715730
##         [,17]    [,18]    [,19]    [,20]    [,21]    [,22]    [,23]    [,24]
## [1,] 1.737537 1.700033 1.729419 1.723374 1.694986 1.641363 1.696617 1.698210
## [2,] 2.072888 2.039289 2.065615 2.060200 2.034768 1.986730 2.036230 2.027861
## [3,] 2.722986 2.696959 2.717352 2.713157 2.693457 2.656244 2.694589 2.657633
##         [,25]    [,26]    [,27]    [,28]    [,29]    [,30]    [,31]    [,32]
## [1,] 1.698068 1.601066 1.674596 1.654408 1.713730 1.743856 1.641478 1.687953
## [2,] 2.027735 1.925966 1.990218 1.971489 2.048950 2.057318 1.969307 2.009260
## [3,] 2.657537 2.530962 2.577939 2.561926 2.673163 2.625877 2.563925 2.592048
##         [,33]    [,34]    [,35]    [,36]    [,37]    [,38]    [,39]    [,40]
## [1,] 1.667943 1.654198 1.646068 1.725356 1.780965 1.755475 1.596308 1.462103
## [2,] 1.992058 1.980242 1.973253 2.041414 2.196226 2.151582 1.938782 1.757781
## [3,] 2.579940 2.571622 2.566702 2.614682 3.069684 2.984750 2.659142 2.379708
##         [,41]    [,42]    [,43]    [,44]    [,45]    [,46]    [,47]    [,48]
## [1,] 1.478617 1.505761 1.515575 1.479539 1.610272 1.653519 1.737130 1.689464
## [2,] 1.773335 1.798900 1.809427 1.767150 2.030655 2.089615 2.202002 2.108515
## [3,] 2.393242 2.415487 2.427513 2.372109 3.136269 3.257321 3.391312 3.117503
##         [,49]    [,50]    [,51]    [,52]    [,53]    [,54]    [,55]    [,56]
## [1,] 1.684556 1.740972 1.785226 1.797470 1.687451 1.647426 1.692116 1.684705
## [2,] 2.103498 2.161158 2.249797 2.262741 2.074766 2.034969 2.079404 2.072035
## [3,] 3.112225 3.172881 3.433498 3.448225 2.965541 2.926272 2.970117 2.962846
##         [,57]    [,58]    [,59]    [,60]    [,61]    [,62]    [,63]    [,64]
## [1,] 1.712829 1.726774 1.742481 1.717602 1.742466 1.727169 1.694157 1.733462
## [2,] 2.101673 2.115558 2.131197 2.106426 2.131183 2.115952 2.083081 2.122217
## [3,] 2.998109 3.011856 3.027340 3.002815 3.027326 3.012246 2.979703 3.018449
##         [,65]    [,66]    [,67]    [,68]    [,69]    [,70]    [,71]    [,72]
## [1,] 1.701567 1.745462 1.731029 1.670456 1.695758 1.726803 1.649632 1.670162
## [2,] 2.090459 2.134166 2.119795 2.059482 2.084676 2.115587 2.018919 2.039361
## [3,] 2.987007 3.030279 3.016051 2.956338 2.981281 3.011885 2.870270 2.890508
##         [,73]    [,74]    [,75]
## [1,] 1.671362 1.558693 1.771397
## [2,] 2.040556 1.825105 2.118379
## [3,] 2.891691 2.338960 2.787640
##
## attr(,"class")
## [1] "mixgpd_predict"
```

```r
# Predict density on a grid
y_grid <- seq(min(y), max(y) * 1.5, length.out = 200)
pred_d <- predict(fit, y = y_grid, type = "density")
pred_d


## $fit
##     id         y     density        lower       upper
## 1    1 0.2011054 0.467657195 3.591474e-01 0.576721770
## 2    1 0.2197050 0.519768592 4.130050e-01 0.632806362
## 3    1 0.2383046 0.573240746 4.703176e-01 0.687472302
## 4    1 0.2569042 0.627284254 5.299236e-01 0.741241233
## 5    1 0.2755038 0.681007030 5.907134e-01 0.793479276
## 6    1 0.2941034 0.733435540 6.514513e-01 0.841575663
## 7    1 0.3127030 0.783542117 7.062802e-01 0.886530092
## 8    1 0.3313026 0.830277438 7.514943e-01 0.936967664
## 9    1 0.3499022 0.872606845 7.950619e-01 0.979253545
## 10   1 0.3685018 0.909548793 8.201120e-01 1.015551155
## 11   1 0.3871014 0.940213446 8.403055e-01 1.054872247
## 12   1 0.4057010 0.963839259 8.552549e-01 1.089643036
## 13   1 0.4243006 0.979825425 8.646758e-01 1.113955373
## 14   1 0.4429002 0.987758208 8.683968e-01 1.127063868
## 15   1 0.4614998 0.987429516 8.663651e-01 1.128562624
## 16   1 0.4800994 0.978846514 8.586489e-01 1.118406273
## 17   1 0.4986990 0.962231654 8.451759e-01 1.096912349
## 18   1 0.5172986 0.938013105 8.258833e-01 1.064744784
## 19   1 0.5358982 0.906806155 8.009708e-01 1.031215894
## 20   1 0.5544978 0.869386779 7.701384e-01 0.996191195
## 21   1 0.5730974 0.826658981 7.353758e-01 0.950669784
## 22   1 0.5916970 0.779617890 6.973369e-01 0.898302890
## 23   1 0.6102966 0.729310767 6.567109e-01 0.841917601
## 24   1 0.6288963 0.676798087 6.058733e-01 0.792794182
## 25   1 0.6474959 0.623116729 5.449757e-01 0.743084279
## 26   1 0.6660955 0.569247031 4.855416e-01 0.692294078
## 27   1 0.6846951 0.516085092 4.274739e-01 0.641203809
## 28   1 0.7032947 0.464421249 3.723367e-01 0.590539636
## 29   1 0.7218943 0.437176460 3.208527e-01 0.962572636
## 30   1 0.7404939 0.598347085 2.735393e-01 1.022443922
## 31   1 0.7590935 0.860189722 6.329170e-01 1.018319603
## 32   1 0.7776931 0.834755039 6.199054e-01 0.982938234
## 33   1 0.7962927 0.810064079 6.070879e-01 0.949260499
## 34   1 0.8148923 0.786094100 5.944625e-01 0.916218851
## 35   1 0.8334919 0.762823126 5.820275e-01 0.884855149
## 36   1 0.8520915 0.740229920 5.697812e-01 0.855243681
## 37   1 0.8706911 0.718293954 5.577218e-01 0.826617481
## 38   1 0.8892907 0.696995383 5.458474e-01 0.799363393
## 39   1 0.9078903 0.676315019 5.340936e-01 0.773735682
```

```
## 40   1 0.9264899 0.656234307 5.217663e-01 0.748924628
## 41   1 0.9450895 0.636735301 5.096548e-01 0.724904364
## 42   1 0.9636891 0.617800642 4.977565e-01 0.701649841
## 43   1 0.9822887 0.599413536 4.860689e-01 0.679136795
## 44   1 1.0008883 0.581557734 4.745895e-01 0.657341728
## 45   1 1.0194879 0.564217511 4.633157e-01 0.636241882
## 46   1 1.0380876 0.547377647 4.522450e-01 0.615815217
## 47   1 1.0566872 0.531023411 4.413751e-01 0.596040385
## 48   1 1.0752868 0.515140542 4.309394e-01 0.576896712
## 49   1 1.0938864 0.499715230 4.210056e-01 0.558364177
## 50   1 1.1124860 0.484734105 4.112396e-01 0.540423388
## 51   1 1.1310856 0.470184217 4.016395e-01 0.523055566
## 52   1 1.1496852 0.456053023 3.922036e-01 0.506242524
## 53   1 1.1682848 0.442328371 3.829920e-01 0.489966648
## 54   1 1.1868844 0.428998491 3.739814e-01 0.474210881
## 55   1 1.2054840 0.416051976 3.651247e-01 0.458958703
## 56   1 1.2240836 0.403477774 3.559181e-01 0.444194117
## 57   1 1.2426832 0.391265172 3.467293e-01 0.429901630
## 58   1 1.2612828 0.379403787 3.377314e-01 0.416066239
## 59   1 1.2798824 0.367883553 3.283629e-01 0.403757793
## 60   1 1.2984820 0.356694714 3.177328e-01 0.391810454
## 61   1 1.3170816 0.345827806 3.075237e-01 0.379638781
## 62   1 1.3356812 0.335273657 2.988533e-01 0.367836147
## 63   1 1.3542808 0.325023367 2.903818e-01 0.356806873
## 64   1 1.3728804 0.315068306 2.821055e-01 0.347226921
## 65   1 1.3914800 0.305400105 2.718413e-01 0.337356936
## 66   1 1.4100796 0.296010641 2.616519e-01 0.327717756
## 67   1 1.4286792 0.286892036 2.518095e-01 0.318305184
## 68   1 1.4472789 0.278036645 2.423035e-01 0.308983327
## 69   1 1.4658785 0.269437050 2.331432e-01 0.300756781
## 70   1 1.4844781 0.261086050 2.243638e-01 0.292432131
## 71   1 1.5030777 0.252976658 2.158929e-01 0.284292735
## 72   1 1.5216773 0.245102091 2.077203e-01 0.276470941
## 73   1 1.5402769 0.237455764 1.998363e-01 0.269011156
## 74   1 1.5588765 0.230031285 1.922314e-01 0.261711779
## 75   1 1.5774761 0.222822446 1.848964e-01 0.254570237
## 76   1 1.5960757 0.215823221 1.778225e-01 0.247583989
## 77   1 1.6146753 0.209027755 1.710009e-01 0.240750518
## 78   1 1.6332749 0.202430365 1.644234e-01 0.234067337
## 79   1 1.6518745 0.196025530 1.582131e-01 0.227531985
## 80   1 1.6704741 0.189807886 1.526611e-01 0.221142026
## 81   1 1.6890737 0.183772222 1.473159e-01 0.214895053
## 82   1 1.7076733 0.177913479 1.417062e-01 0.208788685
## 83   1 1.7262729 0.172226736 1.359325e-01 0.202820566
## 84   1 1.7448725 0.166707216 1.303726e-01 0.196988368
## 85   1 1.7634721 0.161350275 1.250194e-01 0.191289788
```

```
## 86   1 1.7820717 0.156151400 1.198660e-01 0.185722550
## 87   1 1.8006713 0.151106204 1.149058e-01 0.180194648
## 88   1 1.8192709 0.146210424 1.101321e-01 0.174763570
## 89   1 1.8378705 0.141459917 1.055387e-01 0.169453456
## 90   1 1.8564701 0.136850652 1.011195e-01 0.164262574
## 91   1 1.8750698 0.132378713 9.686848e-02 0.159189204
## 92   1 1.8936694 0.128040293 9.277994e-02 0.154231637
## 93   1 1.9122690 0.123831688 8.884828e-02 0.149388176
## 94   1 1.9308686 0.119749298 8.506809e-02 0.144657135
## 95   1 1.9494682 0.115789622 8.144532e-02 0.140036838
## 96   1 1.9680678 0.111949253 7.791805e-02 0.135525622
## 97   1 1.9866674 0.108224881 7.448012e-02 0.131121835
## 98   1 2.0052670 0.104613283 7.117787e-02 0.126823834
## 99   1 2.0238666 0.101111326 6.800660e-02 0.122629990
## 100  1 2.0424662 0.097715962 6.496175e-02 0.118538685
## 101  1 2.0610658 0.094424225 6.207887e-02 0.114548310
## 102  1 2.0796654 0.091233231 5.934950e-02 0.110657269
## 103  1 2.0982650 0.088140172 5.672924e-02 0.106863978
## 104  1 2.1168646 0.085142317 5.421417e-02 0.103166862
## 105  1 2.1354642 0.082237009 5.180049e-02 0.099564359
## 106  1 2.1540638 0.079421660 4.948453e-02 0.096045209
## 107  1 2.1726634 0.076693755 4.726273e-02 0.092580131
## 108  1 2.1912630 0.074050844 4.513164e-02 0.089206105
## 109  1 2.2098626 0.071490544 4.308793e-02 0.085921648
## 110  1 2.2284622 0.069010533 4.112839e-02 0.082725289
## 111  1 2.2470618 0.066608554 3.924989e-02 0.079805614
## 112  1 2.2656614 0.064282408 3.744941e-02 0.077376170
## 113  1 2.2842611 0.062029955 3.572405e-02 0.075011436
## 114  1 2.3028607 0.059849112 3.407098e-02 0.072709967
## 115  1 2.3214603 0.057737851 3.248749e-02 0.070271546
## 116  1 2.3400599 0.055694198 3.097092e-02 0.067592579
## 117  1 2.3586595 0.053716231 2.951875e-02 0.064990400
## 118  1 2.3772591 0.051802077 2.812851e-02 0.062463584
## 119  1 2.3958587 0.049949915 2.679782e-02 0.060231322
## 120  1 2.4144583 0.048157970 2.552439e-02 0.058355828
## 121  1 2.4330579 0.046424515 2.430601e-02 0.056504016
## 122  1 2.4516575 0.044747867 2.314052e-02 0.054811695
## 123  1 2.4702571 0.043126388 2.202587e-02 0.053170227
## 124  1 2.4888567 0.041558483 2.097033e-02 0.051578089
## 125  1 2.5074563 0.040042598 1.997189e-02 0.050033808
## 126  1 2.5260559 0.038577220 1.901669e-02 0.048535955
## 127  1 2.5446555 0.037160877 1.810305e-02 0.046965970
## 128  1 2.5632551 0.035792132 1.722934e-02 0.045656509
## 129  1 2.5818547 0.034469589 1.632558e-02 0.044386059
## 130  1 2.6004543 0.033191886 1.543987e-02 0.043153383
## 131  1 2.6190539 0.031957699 1.463270e-02 0.041957287
```

```
## 132  1 2.6376535 0.030765736 1.386337e-02 0.040796615
## 133  1 2.6562531 0.029614741 1.313033e-02 0.039656768
## 134  1 2.6748527 0.028503488 1.243209e-02 0.038540669
## 135  1 2.6934524 0.027430785 1.176720e-02 0.037457631
## 136  1 2.7120520 0.026395470 1.113428e-02 0.036406627
## 137  1 2.7306516 0.025396413 1.053199e-02 0.035386660
## 138  1 2.7492512 0.024432512 9.867364e-03 0.034396769
## 139  1 2.7678508 0.023502694 9.098833e-03 0.033436024
## 140  1 2.7864504 0.022605914 8.374812e-03 0.032486305
## 141  1 2.8050500 0.021741154 7.693658e-03 0.031551677
## 142  1 2.8236496 0.020907424 7.053751e-03 0.030569907
## 143  1 2.8422492 0.020103758 6.453498e-03 0.029616452
## 144  1 2.8608488 0.019329217 5.891330e-03 0.028796977
## 145  1 2.8794484 0.018582885 5.365706e-03 0.028020033
## 146  1 2.8980480 0.017863871 4.875109e-03 0.027265488
## 147  1 2.9166476 0.017171307 4.418048e-03 0.026532656
## 148  1 2.9352472 0.016504349 3.993059e-03 0.025820873
## 149  1 2.9538468 0.015862174 3.640959e-03 0.025129495
## 150  1 2.9724464 0.015243982 3.323954e-03 0.024457900
## 151  1 2.9910460 0.014648992 3.029233e-03 0.023805485
## 152  1 3.0096456 0.014076446 2.755635e-03 0.023159868
## 153  1 3.0282452 0.013525604 2.502037e-03 0.022526338
## 154  1 3.0468448 0.012995749 2.267356e-03 0.021911049
## 155  1 3.0654444 0.012486179 2.050544e-03 0.021313451
## 156  1 3.0840440 0.011996215 1.850591e-03 0.020733009
## 157  1 3.1026436 0.011525193 1.666522e-03 0.020169206
## 158  1 3.1212433 0.011072468 1.497399e-03 0.019621541
## 159  1 3.1398429 0.010637413 1.342318e-03 0.019089531
## 160  1 3.1584425 0.010219419 1.200408e-03 0.018572704
## 161  1 3.1770421 0.009817890 1.070833e-03 0.018070607
## 162  1 3.1956417 0.009432251 9.527902e-04 0.017582798
## 163  1 3.2142413 0.009061939 8.455080e-04 0.017108851
## 164  1 3.2328409 0.008706409 7.482477e-04 0.016648353
## 165  1 3.2514405 0.008365131 6.603015e-04 0.016200902
## 166  1 3.2700401 0.008037588 5.809925e-04 0.015766111
## 167  1 3.2886397 0.007723279 5.104434e-04 0.015343604
## 168  1 3.3072393 0.007421718 4.488611e-04 0.014933016
## 169  1 3.3258389 0.007132431 3.934252e-04 0.014533995
## 170  1 3.3444385 0.006854959 3.436568e-04 0.014146198
## 171  1 3.3630381 0.006588857 2.991035e-04 0.013769294
## 172  1 3.3816377 0.006333690 2.593380e-04 0.013402962
## 173  1 3.4002373 0.006089040 2.239581e-04 0.013046891
## 174  1 3.4188369 0.005854498 1.925851e-04 0.012700779
## 175  1 3.4374365 0.005629670 1.648634e-04 0.012364333
## 176  1 3.4560361 0.005414171 1.404599e-04 0.012037270
## 177  1 3.4746357 0.005207631 1.190629e-04 0.011719316
```

```
## 178  1 3.4932353 0.005009691 1.003812e-04 0.011410204
## 179  1 3.5118349 0.004820002 8.414391e-05 0.011109676
## 180  1 3.5304346 0.004638228 7.009919e-05 0.010817481
## 181  1 3.5490342 0.004464043 5.801362e-05 0.010533378
## 182  1 3.5676338 0.004297133 4.767150e-05 0.010257131
## 183  1 3.5862334 0.004137198 3.887403e-05 0.009988512
## 184  1 3.6048330 0.003983945 3.143860e-05 0.009727300
## 185  1 3.6234326 0.003837093 2.519802e-05 0.009473280
## 186  1 3.6420322 0.003696368 1.999982e-05 0.009226244
## 187  1 3.6606318 0.003561508 1.570549e-05 0.008985992
## 188  1 3.6792314 0.003432258 1.218976e-05 0.008752327
## 189  1 3.6978310 0.003308372 9.339865e-06 0.008525060
## 190  1 3.7164306 0.003189612 7.054857e-06 0.008304007
## 191  1 3.7350302 0.003075751 5.244881e-06 0.008088989
## 192  1 3.7536298 0.002966567 3.830472e-06 0.007879834
## 193  1 3.7722294 0.002861849 2.741866e-06 0.007676373
## 194  1 3.7908290 0.002761392 1.918306e-06 0.007478445
## 195  1 3.8094286 0.002665000 1.307368e-06 0.007289946
## 196  1 3.8280282 0.002572485 8.642861e-07 0.007113855
## 197  1 3.8466278 0.002483666 5.512898e-07 0.006942335
## 198  1 3.8652274 0.002398371 3.369523e-07 0.006775260
## 199  1 3.8838270 0.002316434 1.955478e-07 0.006612507
## 200  1 3.9024266 0.002237699 1.064203e-07 0.006453955
##
## $lower
## NULL
##
## $upper
## NULL
##
## $type
## [1] "density"
##
## $grid
##   [1] 0.2011054 0.2197050 0.2383046 0.2569042 0.2755038 0.2941034 0.3127030
##   [8] 0.3313026 0.3499022 0.3685018 0.3871014 0.4057010 0.4243006 0.4429002
##  [15] 0.4614998 0.4800994 0.4986990 0.5172986 0.5358982 0.5544978 0.5730974
##  [22] 0.5916970 0.6102966 0.6288963 0.6474959 0.6660955 0.6846951 0.7032947
##  [29] 0.7218943 0.7404939 0.7590935 0.7776931 0.7962927 0.8148923 0.8334919
##  [36] 0.8520915 0.8706911 0.8892907 0.9078903 0.9264899 0.9450895 0.9636891
##  [43] 0.9822887 1.0008883 1.0194879 1.0380876 1.0566872 1.0752868 1.0938864
##  [50] 1.1124860 1.1310856 1.1496852 1.1682848 1.1868844 1.2054840 1.2240836
##  [57] 1.2426832 1.2612828 1.2798824 1.2984820 1.3170816 1.3356812 1.3542808
##  [64] 1.3728804 1.3914800 1.4100796 1.4286792 1.4472789 1.4658785 1.4844781
##  [71] 1.5030777 1.5216773 1.5402769 1.5588765 1.5774761 1.5960757 1.6146753
##  [78] 1.6332749 1.6518745 1.6704741 1.6890737 1.7076733 1.7262729 1.7448725
```

```
##  [85] 1.7634721 1.7820717 1.8006713 1.8192709 1.8378705 1.8564701 1.8750698
##  [92] 1.8936694 1.9122690 1.9308686 1.9494682 1.9680678 1.9866674 2.0052670
##  [99] 2.0238666 2.0424662 2.0610658 2.0796654 2.0982650 2.1168646 2.1354642
## [106] 2.1540638 2.1726634 2.1912630 2.2098626 2.2284622 2.2470618 2.2656614
## [113] 2.2842611 2.3028607 2.3214603 2.3400599 2.3586595 2.3772591 2.3958587
## [120] 2.4144583 2.4330579 2.4516575 2.4702571 2.4888567 2.5074563 2.5260559
## [127] 2.5446555 2.5632551 2.5818547 2.6004543 2.6190539 2.6376535 2.6562531
## [134] 2.6748527 2.6934524 2.7120520 2.7306516 2.7492512 2.7678508 2.7864504
## [141] 2.8050500 2.8236496 2.8422492 2.8608488 2.8794484 2.8980480 2.9166476
## [148] 2.9352472 2.9538468 2.9724464 2.9910460 3.0096456 3.0282452 3.0468448
## [155] 3.0654444 3.0840440 3.1026436 3.1212433 3.1398429 3.1584425 3.1770421
## [162] 3.1956417 3.2142413 3.2328409 3.2514405 3.2700401 3.2886397 3.3072393
## [169] 3.3258389 3.3444385 3.3630381 3.3816377 3.4002373 3.4188369 3.4374365
## [176] 3.4560361 3.4746357 3.4932353 3.5118349 3.5304346 3.5490342 3.5676338
## [183] 3.5862334 3.6048330 3.6234326 3.6420322 3.6606318 3.6792314 3.6978310
## [190] 3.7164306 3.7350302 3.7536298 3.7722294 3.7908290 3.8094286 3.8280282
## [197] 3.8466278 3.8652274 3.8838270 3.9024266
##
## attr(,"class")
## [1] "mixgpd_predict"
```

5. Real data application: conditional ozone modeling

To provide a fully reproducible illustration without external files, we analyze the built-in `airquality` dataset in base R, which records daily ozone and meteorological measurements in New York during 1973. Ozone concentrations are right-skewed with occasional high episodes. While not an extreme-value benchmark dataset, it demonstrates conditional distribution modeling and high-quantile inference in a familiar setting.

5.1. Data preprocessing

```r
dat <- airquality
dat <- dat[complete.cases(dat[, c("Ozone", "Solar.R", "Wind", "Temp")]), ]

y <- as.numeric(dat$Ozone)

X <- data.frame(
  Solar.R = dat$Solar.R,
  Wind    = dat$Wind,
  Temp    = dat$Temp
)

# Scale covariates for numerical stability in MCMC
X_scaled <- as.data.frame(scale(X))
```

```
summary(y)

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1.0    18.0    31.0    42.1    62.0   168.0
```

5.2. Model fitting: bulk-only vs bulk+GPD

We compare a conditional bulk-only DPM model to a conditional bulk+GPD model using the same bulk kernel. For strictly positive outcomes with skewness, a lognormal kernel is a practical default; users may also consider gamma or inverse Gaussian depending on domain knowledge.

```
mcmc <- list(niter = 600, nburnin = 200, thin = 4, nchains = 1, seed = 1)

bundle_bulk <- build_nimble_bundle(
  y = y,
  X = X_scaled,
  backend = "sb",
  kernel = "lognormal",
  GPD = FALSE,
  components = 5,
  mcmc = mcmc
)
fit_bulk <- run_mcmc_bundle_manual(bundle_bulk, show_progress = FALSE)

## ===== Monitors =====
## thin = 1: alpha, beta_meanlog, sdlog, w, z
## ===== Samplers =====
## RW sampler (20)
##   - alpha
##   - beta_meanlog[]  (15 elements)
##   - v[]  (4 elements)
## conjugate sampler (5)
##   - sdlog[]  (5 elements)
## categorical sampler (111)
##   - z[]  (111 elements)

bundle_gpd <- build_nimble_bundle(
  y = y,
  X = X_scaled,
  backend = "sb",
  kernel = "lognormal",
  GPD = TRUE,
  components = 4,
  mcmc = mcmc
```

```
)
fit_gpd <- run_mcmc_bundle_manual(bundle_gpd, show_progress = FALSE)

## ===== Monitors =====
## thin = 1: alpha, beta_meanlog, beta_tail_scale, beta_threshold, sdlog, sdlog_u, tail_sh
## ===== Samplers =====
## RW sampler (136)
##   - alpha
##   - sdlog[]  (4 elements)
##   - beta_meanlog[]  (12 elements)
##   - sdlog_u
##   - beta_tail_scale[]  (3 elements)
##   - tail_shape
##   - v[]  (3 elements)
##   - threshold[]  (111 elements)
## conjugate sampler (3)
##   - beta_threshold[]  (3 elements)
## categorical sampler (111)
##   - z[]  (111 elements)

print(fit_bulk)

## MixGPD fit | backend: Stick-Breaking Process | kernel: Lognormal Distribution | GPD tai
## n = 111 | components = 5 | epsilon = 0.025
## MCMC: niter=600, nburnin=200, thin=4, nchains=1
## Fit
## Use summary() for posterior summaries; plot() for diagnostics; predict() for prediction

print(fit_gpd)

## MixGPD fit | backend: Stick-Breaking Process | kernel: Lognormal Distribution | GPD tai
## n = 111 | components = 4 | epsilon = 0.025
## MCMC: niter=600, nburnin=200, thin=4, nchains=1
## Fit
## Use summary() for posterior summaries; plot() for diagnostics; predict() for prediction
```

5.3. Conditional high-quantile prediction

We compute conditional 0.90, 0.95, and 0.99 quantiles under three representative covariate profiles: low temperature/high wind, average conditions, and high temperature/low wind.

```
X_prof <- rbind(
  low_ozone    = c(Solar.R = -0.5, Wind =  1.0, Temp = -1.0),
  average     = c(Solar.R =  0.0, Wind =  0.0, Temp =  0.0),
  high_ozone   = c(Solar.R =  1.0, Wind = -1.0, Temp =  1.0)
```

```
)
X_prof <- as.data.frame(X_prof)

taus <- c(0.90, 0.95, 0.99)

pred_bulk_q <- predict(fit_bulk, newdata = X_prof, type = "quantile", index = taus)
pred_gpd_q  <- predict(fit_gpd,  newdata = X_prof, type = "quantile", index = taus)

pred_bulk_q

## $fit
##      estimate index id     lower        upper
## 1   38.712555  0.90  1 0.5948691   188.176213
## 2    2.142936  0.90  2 1.3302706     6.351091
## 3  598.750561  0.90  3 0.4046604  3565.620211
## 4  100.496270  0.95  1 0.7330061  1139.977380
## 5    4.330256  0.95  2 1.5243656    20.564533
## 6  838.360197  0.95  3 0.4279975  4155.646889
## 7 2099.367108  0.99  1 1.0855245  7485.157515
## 8   52.633765  0.99  2 2.1039323   427.162311
## 9 2543.020113  0.99  3 0.8249382 22465.187879
##
## $lower
## NULL
##
## $upper
## NULL
##
## $type
## [1] "quantile"
##
## $grid
## [1] 0.90 0.95 0.99
##
## $draws
## , , 1
##
##                [,1]     [,2]         [,3]
##   [1,] 1265.4535158 2.482378 8.119003e-01
##   [2,] 1119.8810465 2.618210 4.091251e-01
##   [3,]   18.6719347 1.867054 4.006208e-01
##   [4,]   10.5947508 1.329570 2.177078e-01
##   [5,]   70.0467743 1.747179 4.404585e-01
##   [6,]   25.5857168 1.697823 2.344597e-01
##   [7,]    1.4050094 1.518107 1.044637e+02
##   [8,]    2.1999032 1.417171 5.355931e+02
```

```
##   [9,]    5.9770364 1.408715 1.694641e+03
##  [10,]    1.2855233 1.698498 1.587663e+04
##  [11,]    2.0805621 1.606113 1.007905e+04
##  [12,]    1.1939839 1.329182 5.092390e+03
##  [13,]    1.0854166 1.673313 2.057491e+02
##  [14,]    0.5680540 1.381666 4.792449e+01
##  [15,]    1.1392466 2.312427 1.009327e+02
##  [16,]    2.6214768 1.701674 1.428539e+02
##  [17,]    2.2342333 1.406116 1.169602e+02
##  [18,]    0.8154500 2.072621 1.532798e+02
##  [19,]   10.6131472 1.499244 3.137638e+02
##  [20,]    0.9929753 1.918877 3.998611e+02
##  [21,]    1.4771946 1.734589 1.520619e+02
##  [22,]    1.1449656 1.605614 2.933929e+02
##  [23,]    0.6986267 1.444777 2.792368e+02
##  [24,]    1.6156789 1.352813 3.398821e+01
##  [25,]  164.4457682 1.831193 8.860249e+01
##  [26,]   49.1103535 1.407497 1.348131e+02
##  [27,]    0.7159994 1.467733 7.571331e+01
##  [28,]    1.3391727 1.526923 2.605094e+02
##  [29,]    0.7355001 2.231912 2.867995e+02
##  [30,]    2.7154340 1.433802 1.479038e+02
##  [31,]   40.8550174 1.331045 1.211927e+02
##  [32,]    6.2233449 1.338263 7.766488e+01
##  [33,]   20.7601505 2.554543 1.896527e+02
##  [34,]   49.2011342 3.640570 1.685476e+02
##  [35,]  209.6466146 6.541089 1.627268e+02
##  [36,]   69.7392857 6.276067 1.684399e+02
##  [37,]   59.8552629 6.418969 2.839740e+02
##  [38,]   69.4063175 2.862655 1.701151e+02
##  [39,]   51.1017961 8.289066 2.193637e+02
##  [40,]   37.0730060 2.717245 1.627029e+02
##  [41,]   53.3499040 1.770479 6.881450e+01
##  [42,]   45.9422824 1.488970 1.018519e+02
##  [43,]   22.5532069 3.136027 1.866045e+02
##  [44,]   24.7051129 2.013330 2.388696e+02
##  [45,]   28.1640542 1.630369 3.976512e+02
##  [46,]   21.1600951 2.086144 3.376158e+02
##  [47,]   25.2949624 1.864578 1.318468e+02
##  [48,]   49.2758185 3.175120 2.075014e+02
##  [49,]   21.3636219 2.433450 1.342336e+02
##  [50,]   60.0868257 4.681922 1.987638e+02
##  [51,]   13.2681343 2.331691 1.323131e+02
##  [52,]    9.9460862 1.452102 1.095516e+02
##  [53,]    2.2991611 2.682987 5.977792e+01
##  [54,]    5.9180260 2.251284 4.300639e+01
```

```
##  [55,]    1.3128522 1.524832 1.070414e+01
##  [56,]    0.9419972 3.074202 1.403549e+02
##  [57,]    0.9834004 3.767507 1.470980e+02
##  [58,]    0.4852876 1.762914 1.462503e+02
##  [59,]    0.7962113 2.416537 3.814926e+02
##  [60,]    2.1420060 2.298964 1.445570e+02
##  [61,]    0.9732818 2.426207 1.534965e+02
##  [62,]    3.6388487 1.207854 2.631526e+02
##  [63,]    6.5771407 1.559468 1.619356e+02
##  [64,]    0.8592890 1.880152 3.832220e+02
##  [65,]    0.7361008 1.580770 1.733714e+02
##  [66,]    1.9996073 1.904064 2.012182e+02
##  [67,]    3.2752767 2.548615 1.982442e+02
##  [68,]    1.4619493 1.752950 5.690220e+01
##  [69,]    3.0339422 1.723932 1.458368e+02
##  [70,]    3.4557875 2.561887 1.797510e+02
##  [71,]    3.7973258 2.494940 4.767500e+02
##  [72,]    4.9910441 1.520266 1.281683e+03
##  [73,]   11.5818952 1.458596 5.980617e+02
##  [74,]    1.2184457 1.693618 1.878138e+03
##  [75,]    0.5813252 1.648617 1.469632e+03
##  [76,]    0.8855783 1.748478 5.085428e+02
##  [77,]    1.0130751 3.085243 6.369494e+02
##  [78,]    2.8353706 2.238428 9.066453e+02
##  [79,]    2.5947116 2.571240 1.716150e+03
##  [80,]    1.3222622 2.158197 1.646278e+03
##  [81,]    2.7301643 1.719272 1.486205e+03
##  [82,]    2.1656813 1.968964 1.037835e+03
##  [83,]    1.7321945 1.827476 4.209806e+02
##  [84,]    1.2748921 1.718120 2.217497e+02
##  [85,]    0.6775261 1.532859 9.627484e+01
##  [86,]    0.7138772 1.605003 1.425515e+02
##  [87,]    1.1865942 1.868547 1.084442e+02
##  [88,]    1.0896570 1.681646 4.404506e+01
##  [89,]    1.1187096 1.556045 7.027218e+01
##  [90,]    0.6098386 1.615838 7.438885e+01
##  [91,]    0.9535214 1.459829 7.452150e+01
##  [92,]    1.0763381 1.572972 4.148780e+02
##  [93,]    1.7403210 1.469963 1.435241e+02
##  [94,]    1.1530540 1.698962 1.090420e+02
##  [95,]    1.8167219 1.766633 1.283354e+02
##  [96,]    7.3266780 1.982285 1.040610e+02
##  [97,]    2.5926617 1.756971 3.180953e+01
##  [98,]    3.9661751 1.548838 5.629489e+01
##  [99,]    6.2423468 1.909077 6.942429e+01
## [100,]    1.9609047 1.735032 3.415878e+02
```

```
##
## , , 2
##
##                [,1]      [,2]         [,3]
##   [1,] 1946.0288764  4.449337 8.789255e-01
##   [2,] 1791.6749459  5.097449 4.356529e-01
##   [3,]   28.4268184  2.629681 4.210711e-01
##   [4,]   23.4964422  1.753205 2.323513e-01
##   [5,]   91.1356880  2.327669 4.739711e-01
##   [6,]   33.2025389  2.259121 2.594678e-01
##   [7,]   59.7857499  2.012784 1.256449e+02
##   [8,]    2.5837519  1.678644 6.349296e+02
##   [9,]    7.1689600  1.630238 1.950480e+03
##  [10,]    4.3282277  2.337799 2.041898e+04
##  [11,]   88.8691111  1.998510 1.150547e+04
##  [12,]   23.6626471  1.480267 5.510092e+03
##  [13,]   47.4383968  2.612063 2.658340e+02
##  [14,]   19.9408648  1.624768 5.560543e+01
##  [15,]    1.2499308  3.691642 1.610705e+02
##  [16,]    3.3706437  2.494537 1.915439e+02
##  [17,]   25.0497300  1.641890 1.234934e+02
##  [18,]   96.2526567  4.079279 2.408389e+02
##  [19,]   11.4480862  2.447292 5.125742e+02
##  [20,]    7.5338453  2.967541 6.371865e+02
##  [21,]    1.5587249  2.469771 2.296830e+02
##  [22,]    1.2446934  2.217145 3.947981e+02
##  [23,]    0.7498146  1.955731 4.030563e+02
##  [24,]    2.1912575  1.493382 3.781852e+01
##  [25,]  419.6800699  2.859996 1.448133e+02
##  [26,]   70.9225238  1.620582 1.666424e+02
##  [27,]    0.8254632  1.758597 9.059492e+01
##  [28,]    5.5082940  2.323502 3.056449e+02
##  [29,]    1.7668887  4.040796 4.429347e+02
##  [30,]    2.8799762  2.040056 2.278153e+02
##  [31,]   50.0602387  1.501278 1.490768e+02
##  [32,]    8.4763469  1.594385 9.485607e+01
##  [33,]   80.2569071  4.774539 3.304233e+02
##  [34,]  225.3152662  8.605283 4.046775e+02
##  [35,] 2914.4748938 28.961428 5.056763e+02
##  [36,]  393.0914000 19.835205 7.359308e+02
##  [37,]  163.4022467 17.298672 1.283733e+03
##  [38,]  141.1422750  5.840174 5.261597e+02
##  [39,]  151.6973725 28.119388 1.434032e+03
##  [40,]   51.8931086  6.277619 4.533864e+02
##  [41,]   73.3639153  2.663462 1.285663e+02
##  [42,]   76.9890885  1.897836 1.248263e+02
```

```
##  [43,]   45.8883293  5.380319 3.760333e+02
##  [44,]   45.0168538  2.987467 3.107878e+02
##  [45,]   35.6149901  2.041724 5.570385e+02
##  [46,]   33.6706688  3.147293 6.204343e+02
##  [47,]   32.3270649  2.552722 1.447793e+02
##  [48,]  154.7771284  6.474756 3.661861e+02
##  [49,]   40.8319557  3.950938 2.590562e+02
##  [50,]  194.6572520 10.732621 5.739319e+02
##  [51,]   22.4344376  5.204061 4.215467e+02
##  [52,]   12.3809441  2.723042 2.917881e+02
##  [53,]   37.6345653 10.009665 2.637051e+02
##  [54,]   33.6066597  6.832868 2.716120e+02
##  [55,]    2.5963225  4.667368 3.858612e+02
##  [56,]    1.1850925 21.224401 9.583937e+02
##  [57,]    1.4126725 15.121251 5.925491e+02
##  [58,]    0.5056957  2.870116 2.336123e+02
##  [59,]    0.8524034  5.642801 7.936545e+02
##  [60,]    2.2076045  5.218121 3.486860e+02
##  [61,]    1.0335335  7.319144 1.114454e+03
##  [62,]    3.7667502  1.549883 3.605682e+02
##  [63,]    6.9714560  2.277036 2.775161e+02
##  [64,]    0.9140107  3.211859 6.312851e+02
##  [65,]    0.8164738  2.445828 2.180837e+02
##  [66,]    2.1764194  3.206663 2.702311e+02
##  [67,]    3.4397648  3.838005 3.071849e+02
##  [68,]    1.7067793  2.545377 5.862481e+01
##  [69,]    3.4504346  2.224782 1.904490e+02
##  [70,]    5.7317710  3.984381 1.898537e+02
##  [71,]    4.2841658  3.756471 7.573611e+02
##  [72,]    6.0927246  1.781828 1.407557e+03
##  [73,]   12.3702091  1.856888 7.087119e+02
##  [74,]    1.2737511  2.220478 2.658629e+03
##  [75,]    0.6353773  2.129091 1.997948e+03
##  [76,]    1.5387905  2.424676 7.121743e+02
##  [77,]    3.7581829  6.177361 1.368864e+03
##  [78,]    6.5378997  3.561522 1.458572e+03
##  [79,]    5.9953262  4.250830 2.565757e+03
##  [80,]    1.7841106  2.866151 1.979004e+03
##  [81,]   10.2065949  2.428058 1.605747e+03
##  [82,]   26.3195047  6.476285 1.299292e+03
##  [83,]    2.2134299  2.331054 4.770333e+02
##  [84,]    1.5549764  2.151287 2.533929e+02
##  [85,]    1.5697006  5.396933 1.036642e+02
##  [86,]    0.9318271  2.163645 1.515918e+02
##  [87,]    2.4802113  2.594473 1.227187e+02
##  [88,]    1.4945234  1.959447 5.236339e+01
```

```
##  [89,]    1.2850628  1.880957 7.883632e+01
##  [90,]    0.7177984  2.081539 8.417942e+01
##  [91,]    1.1039660  1.633912 8.380270e+01
##  [92,]    1.2810789  1.931152 4.604382e+02
##  [93,]    2.0114206  1.838271 1.546080e+02
##  [94,]    1.3265943  2.076123 1.331179e+02
##  [95,]    2.2074122  2.237937 1.492826e+02
##  [96,]   23.3404462  2.888417 1.169306e+02
##  [97,]   19.3147355  2.595093 3.548463e+01
##  [98,]   11.3279891  1.869291 6.099681e+01
##  [99,]   10.5870314  2.504533 7.426736e+01
## [100,]    6.3354862  2.214861 3.841985e+02
##
## , , 3
##
##                [,1]       [,2]         [,3]
##   [1,] 4.200858e+03  31.402554 1.136335e+00
##   [2,] 4.259909e+03  51.872697 1.768288e+00
##   [3,] 6.053697e+01   4.783257 4.625260e-01
##   [4,] 3.943470e+02  21.257671 1.146619e+00
##   [5,] 1.431805e+02   4.203208 5.431979e-01
##   [6,] 5.151735e+01   4.059881 4.541375e-01
##   [7,] 6.667741e+03  42.647592 1.725276e+02
##   [8,] 8.127905e+02   2.416408 8.530974e+02
##   [9,] 1.647787e+04  10.471241 2.492831e+03
##  [10,] 4.233159e+02  12.474417 3.152736e+04
##  [11,] 2.207297e+03  20.090853 1.441379e+04
##  [12,] 6.832723e+01   2.464360 6.304056e+03
##  [13,] 7.412089e+03 135.037536 4.125126e+02
##  [14,] 7.549910e+01   3.305715 7.145975e+01
##  [15,] 1.928419e+00   8.677525 3.556658e+02
##  [16,] 1.465485e+02  15.973242 3.134389e+02
##  [17,] 3.894058e+01   2.175823 1.356425e+02
##  [18,] 7.551267e+03 103.922757 4.986681e+02
##  [19,] 1.235812e+01   5.414495 1.134042e+03
##  [20,] 2.918241e+01   6.319784 1.363584e+03
##  [21,] 1.714661e+00   4.771671 4.560397e+02
##  [22,] 2.383056e+00   4.409630 6.339752e+02
##  [23,] 1.230148e+00   3.303076 6.888876e+02
##  [24,] 3.302009e+00   1.911095 4.510042e+01
##  [25,] 1.355056e+03   6.341704 3.074170e+02
##  [26,] 1.016956e+02   2.212712 2.365193e+02
##  [27,] 3.192584e+00   4.578417 1.253758e+02
##  [28,] 4.406895e+01  17.365198 4.021019e+02
##  [29,] 3.282337e+01  30.526494 9.150711e+02
##  [30,] 3.238809e+00   4.045754 4.523081e+02
```

```
##  [31,] 6.720705e+01   1.940938 2.049280e+02
##  [32,] 1.312621e+01   2.199374 1.293073e+02
##  [33,] 5.090125e+02  18.857296 7.932002e+02
##  [34,] 2.028128e+03  46.615626 1.523263e+03
##  [35,] 1.413729e+05 951.123405 2.704746e+03
##  [36,] 4.810862e+03 144.917474 6.857287e+03
##  [37,] 7.832241e+02 120.349929 1.368106e+04
##  [38,] 3.949350e+02  24.821783 2.920649e+03
##  [39,] 8.369295e+02 351.879742 2.718289e+04
##  [40,] 8.754271e+01  32.595247 2.365607e+03
##  [41,] 1.168662e+02   6.534778 3.387103e+02
##  [42,] 1.623167e+02   3.576287 1.714017e+02
##  [43,] 1.361509e+02  14.126654 1.172600e+03
##  [44,] 1.175277e+02   7.302265 4.767828e+02
##  [45,] 5.160514e+01   3.217298 9.594025e+02
##  [46,] 6.790948e+01   6.781510 1.598954e+03
##  [47,] 4.692436e+01   4.813445 1.695024e+02
##  [48,] 9.129183e+02  31.356767 8.834665e+02
##  [49,] 1.093230e+02   9.224079 7.133017e+02
##  [50,] 1.155972e+03  45.960822 2.983266e+03
##  [51,] 4.550048e+01  29.616419 2.499965e+03
##  [52,] 1.673462e+01  11.193427 1.201516e+03
##  [53,] 1.225247e+03 117.998452 1.891888e+03
##  [54,] 2.777242e+02  40.949684 2.688639e+03
##  [55,] 1.195056e+01 256.375114 2.552912e+04
##  [56,] 2.793433e+01 495.275111 1.907874e+04
##  [57,] 1.646110e+01 151.782995 6.001668e+03
##  [58,] 5.513799e-01   7.215773 4.995099e+02
##  [59,] 2.620363e+00  54.529952 2.793986e+03
##  [60,] 2.351471e+00  20.911006 1.365876e+03
##  [61,] 3.124769e+00  81.990532 1.739891e+04
##  [62,] 4.008588e+00   2.514890 5.850729e+02
##  [63,] 7.742673e+00   4.742723 6.320091e+02
##  [64,] 1.397673e+00  10.176267 1.353606e+03
##  [65,] 4.436148e+00  14.238490 3.161359e+02
##  [66,] 2.490372e+01  98.888406 4.879275e+02
##  [67,] 3.805131e+00   8.146209 6.430799e+02
##  [68,] 2.502275e+00   4.864557 6.167379e+01
##  [69,] 6.494037e+00   3.566499 2.978696e+02
##  [70,] 1.810637e+01  10.670917 2.082438e+02
##  [71,] 1.034751e+01   7.800258 1.660045e+03
##  [72,] 8.610099e+00   2.446866 1.647076e+03
##  [73,] 1.388864e+01   4.223650 9.396627e+02
##  [74,] 1.375830e+00   3.792610 4.746823e+03
##  [75,] 7.626554e-01   3.392767 3.325254e+03
##  [76,] 3.499608e+00   4.554744 1.203265e+03
```

```
##  [77,] 2.212451e+01  21.443664 4.584837e+03
##  [78,] 2.185316e+01   8.663837 3.099401e+03
##  [79,] 2.211168e+01  12.996387 4.937306e+03
##  [80,] 2.999439e+00   5.159316 2.701854e+03
##  [81,] 5.166890e+01   9.445819 1.832603e+03
##  [82,] 7.751590e+02 190.699223 1.887742e+03
##  [83,] 3.438398e+00   3.770695 5.900793e+02
##  [84,] 2.258174e+00   5.476438 3.177860e+02
##  [85,] 2.892907e+02 995.459714 5.294933e+02
##  [86,] 1.340831e+01  63.127917 1.731440e+02
##  [87,] 1.062953e+01   7.909501 1.516242e+02
##  [88,] 2.217506e+00   2.640784 7.012758e+01
##  [89,] 1.642907e+00   3.328829 9.497413e+01
##  [90,] 9.546746e-01   7.130110 1.089781e+02
##  [91,] 1.422011e+00   2.038889 1.023583e+02
##  [92,] 1.771103e+00   4.236591 5.508076e+02
##  [93,] 2.583978e+00   4.822663 1.757135e+02
##  [94,] 1.693741e+00   3.765669 1.857696e+02
##  [95,] 3.179114e+00   5.185036 1.929409e+02
##  [96,] 1.298843e+02  10.058247 1.428444e+02
##  [97,] 4.144319e+02  37.198611 4.296295e+01
##  [98,] 2.007075e+01   2.649179 7.003470e+01
##  [99,] 2.506010e+01   4.109598 8.322218e+01
## [100,] 9.223573e+00   3.570028 4.705427e+02
##
##
## attr(,"class")
## [1] "mixgpd_predict"

pred_gpd_q

## $fit
##      estimate index id        lower       upper
## 1   0.5665563  0.90  1   0.03482007    1.109881
## 2   6.1420268  0.90  2   3.03382007    9.396089
## 3  86.3018850  0.90  3  21.77882536  272.757844
## 4   0.9968649  0.95  1   0.13947459    2.355457
## 5  11.4774844  0.95  2   4.30817258   20.171082
## 6 162.2803702  0.95  3  39.25977188  503.999111
## 7   3.7393406  0.99  1   0.39930006   11.990453
## 8  45.8060418  0.99  2   8.17967962  109.695954
## 9 650.5899695  0.99  3 115.32788000 2109.967800
##
## $lower
## NULL
##
```

```
## $upper
## NULL
##
## $type
## [1] "quantile"
##
## $grid
## [1] 0.90 0.95 0.99
##
## $draws
## , , 1
##
##                [,1]      [,2]      [,3]
##   [1,]  0.171442426  3.060405  48.27346
##   [2,]  0.335995301  3.493047  55.46113
##   [3,]  0.009491249  3.277592  76.55158
##   [4,]  0.234712256  2.588852  56.79373
##   [5,]  0.137224776  2.942396  65.29774
##   [6,]  0.502112859  3.296319  42.82780
##   [7,]  0.202373953  3.208622  58.41924
##   [8,]  0.062815073  3.303199  54.62661
##   [9,]  0.194274770  3.032904  47.36873
##  [10,]  0.459041379  3.587329  65.59952
##  [11,] -0.294676077  3.140949  48.06260
##  [12,]  0.376743417  3.953655  51.45848
##  [13,]  0.119046654  3.953655  48.26685
##  [14,] -0.210985715  3.034832  62.30800
##  [15,]  0.807868097  4.777986  61.56542
##  [16,]  0.170176266  4.777986  74.03986
##  [17,]  0.288238628  3.961855  46.16974
##  [18,]  0.337462427  4.826358  53.95265
##  [19,]  0.394946241  4.826358  30.24852
##  [20,]  0.370020129  5.693126  32.82126
##  [21,]  0.668872772  5.693126  37.23386
##  [22,]  0.572895664  5.693126  31.68666
##  [23,]  0.558422182  5.693126  41.63880
##  [24,]  0.542021851  4.457951  27.04617
##  [25,]  0.336622902  4.457951  36.60935
##  [26,]  0.296824255  4.896146  43.59822
##  [27,]  0.655769872  5.364105  25.40630
##  [28,]  0.447562569  5.364105  27.95226
##  [29,]  0.189119159  5.569745  49.05327
##  [30,]  0.583605343  6.068122  35.39129
##  [31,]  0.696181998  4.859266  54.65774
##  [32,]  0.739529791  5.216173  23.89614
##  [33,]  0.267638160  5.946512  83.15234
```

```
##  [34,]  0.535156377  6.440478  73.58104
##  [35,]  0.664608913  6.440478  32.20241
##  [36,]  0.416986520  5.831038  41.18307
##  [37,]  1.065345081  8.164419  40.61465
##  [38,]  0.247154581  5.396060  33.29955
##  [39,]  0.568525514  5.973210  37.22293
##  [40,]  0.791625577  6.312871  29.36866
##  [41,]  0.822592494  6.312871  41.31636
##  [42,]  1.070850120  7.153829  27.77257
##  [43,]  0.581338666  5.098284  21.43106
##  [44,]  0.456382845  5.211480  31.84841
##  [45,]  0.402778467  4.689400  35.48659
##  [46,]  0.504271944  5.986881  35.12417
##  [47,]  0.884371115  5.986881  22.59652
##  [48,]  0.527470429  4.845341  33.59692
##  [49,]  0.351632161  4.303526  25.95318
##  [50,]  0.460318628  4.868828  41.50372
##  [51,]  0.591734511  5.414104  43.96117
##  [52,]  0.798025159  4.927250  32.26092
##  [53,]  0.632828214  6.331251  51.24787
##  [54,]  0.929337062  6.757773  30.13208
##  [55,]  0.723657250  6.891543  29.17909
##  [56,]  0.981955890  6.841202  27.01908
##  [57,]  0.767506855  8.282498  47.90200
##  [58,]  1.050218864  8.303699  22.75946
##  [59,]  0.680280885  6.390278  22.36798
##  [60,]  1.145194360  6.070320  10.52600
##  [61,]  0.787503143  6.333427  22.16320
##  [62,]  2.042296766  7.456658  12.01295
##  [63,]  0.839668366  9.071581  40.42806
##  [64,]  0.617573809  9.071581  45.51485
##  [65,]  0.889049718  7.567524  27.53078
##  [66,]  0.837066290  7.567524  39.45615
##  [67,]  0.618265611  8.413567  93.48346
##  [68,]  0.415020005  8.350313  46.83813
##  [69,]  0.439512033  8.757528  80.69975
##  [70,]  1.016851658  8.757528  54.53181
##  [71,]  0.791925810  8.250799  35.76207
##  [72,]  1.396883097  8.250799  26.92246
##  [73,]  0.878376715  8.250799  41.11928
##  [74,]  0.705257083 10.543913  69.44839
##  [75,]  0.851933896  8.957393 126.68981
##  [76,]  0.747940414  9.808861 170.16145
##  [77,]  0.327931741  8.078647 202.52057
##  [78,]  0.406170066  7.032746 167.45519
##  [79,]  0.673646136  7.966578 155.35812
```

```
##  [80,]  0.463309778  8.247402 125.06448
##  [81,]  0.843229151  6.071138 162.05064
##  [82,]  0.513026259  7.136099 218.36854
##  [83,]  0.688682010  7.136099 231.73318
##  [84,]  0.684037199  7.136099 229.03317
##  [85,]  0.345891981  7.265396 214.89477
##  [86,]  0.261725482  6.376645 142.56836
##  [87,]  0.422925159  9.689692 346.36152
##  [88,]  0.267814472  5.654265 184.14077
##  [89,]  0.744249357  6.006087 270.80852
##  [90,]  0.287947683  6.919882 237.23731
##  [91,]  0.680256516  6.919882 224.15582
##  [92,]  0.290043867  7.277884 212.53540
##  [93,]  0.320766450  7.380401 143.50922
##  [94,]  0.829937710  7.380401 178.23490
##  [95,]  0.586784204  7.433524 322.70292
##  [96,]  0.697936615  7.433524 207.22662
##  [97,]  0.493185584  7.011070 268.42642
##  [98,]  0.713548253  7.269770 226.42009
##  [99,]  0.616300692  5.498800 233.20507
## [100,]  0.745618254  5.228184 274.52152
##
## , , 2
##
##              [,1]      [,2]      [,3]
##   [1,]  0.2609867  4.358553  69.47699
##   [2,]  0.4170931  5.202711  82.70161
##   [3,]  0.1045679  4.776693 113.20217
##   [4,]  0.2816306  3.492115  76.68651
##   [5,]  0.2024967  4.136283  91.81955
##   [6,]  0.6245816  4.813285  62.72314
##   [7,]  0.2904746  4.642657  85.27897
##   [8,]  0.1780556  4.826750  81.74285
##   [9,]  0.2669547  4.306440  67.29869
##  [10,]  0.5423181  5.392539  98.88157
##  [11,] -0.2021894  4.512266  70.36827
##  [12,]  0.5089601  6.149021  80.15271
##  [13,]  0.3072830  6.149021  76.13672
##  [14,] -0.1457975  4.310087  89.55575
##  [15,]  0.9688854  7.954236 102.99245
##  [16,]  0.3803917  7.954236 124.71036
##  [17,]  0.4873463  6.166290  71.88190
##  [18,]  0.5050633  8.064332  90.16901
##  [19,]  0.7984123  8.064332  51.51406
##  [20,]  0.9021904 10.109604  60.18255
##  [21,]  1.0643486 10.109604  66.34918
```

```
##  [22,]  1.1889463 10.109604  57.51883
##  [23,]  0.8971669 10.109604  74.23231
##  [24,]  0.7978390  7.237187  44.07351
##  [25,]  0.4993106  7.237187  60.02272
##  [26,]  0.5161337  8.223952  73.25304
##  [27,]  1.2058520  9.317450  45.07332
##  [28,]  0.7900391  9.317450  48.59104
##  [29,]  0.5368261  9.810354  88.94005
##  [30,]  1.1450039 11.034875  66.66737
##  [31,]  0.8668997  8.139485  91.93995
##  [32,]  1.3709815  8.967481  42.25555
##  [33,]  0.4739609 10.732251 150.08485
##  [34,]  0.8027014 11.976360 137.07478
##  [35,]  1.1625350 11.976360  59.99358
##  [36,]  0.6751562 10.447161  73.99624
##  [37,]  2.3183560 16.607527  83.57853
##  [38,]  0.6619741  9.393557  60.05542
##  [39,]  0.9280547 10.798482  67.43405
##  [40,]  1.5507926 11.651216  56.31528
##  [41,]  1.2545056 11.651216  76.65369
##  [42,]  2.2707889 13.840220  55.21876
##  [43,]  1.1514420  8.691400  37.72519
##  [44,]  0.7672638  8.956441  54.79628
##  [45,]  0.6080176  7.753769  58.77819
##  [46,]  0.9478385 10.832438  63.57003
##  [47,]  1.7476015 10.832438  41.78764
##  [48,]  0.8395415  8.107660  56.34960
##  [49,]  0.6730621  6.898422  42.77019
##  [50,]  0.6787223  8.161360  69.71957
##  [51,]  0.8280885  9.436614  76.87148
##  [52,]  1.1262326  8.295388  54.56299
##  [53,]  1.0718434 11.697890  94.70691
##  [54,]  1.7926901 12.795851  58.64192
##  [55,]  1.5076932 13.145966  58.40495
##  [56,]  2.0228205 13.013890  52.91403
##  [57,]  1.3794466 16.940057  98.41850
##  [58,]  2.3890243 16.999960  49.11714
##  [59,]  1.4895463 11.848143  43.26759
##  [60,]  2.2496169 11.040368  20.32058
##  [61,]  1.5184814 11.703419  40.95589
##  [62,]  4.1689353 14.654330  24.22635
##  [63,]  1.9000166 19.209130  87.88543
##  [64,]  1.6480229 19.209130  99.78876
##  [65,]  2.1509092 14.955669  55.33941
##  [66,]  1.7424758 14.955669  81.40979
##  [67,]  1.0238055 17.311344 192.77174
```

```
##  [68,]  1.2794432 17.131874 100.88795
##  [69,]  0.8877712 18.296421 171.88519
##  [70,]  1.8551213 18.296421 114.25445
##  [71,]  1.6226533 16.850605  75.29249
##  [72,]  2.9495061 16.850605  56.76983
##  [73,]  1.9536838 16.850605  86.69318
##  [74,]  1.4824772 23.648842 158.84299
##  [75,]  1.2219701 18.875826 267.61968
##  [76,]  1.1424216 21.400016 371.77104
##  [77,]  0.5172459 16.367156 410.33505
##  [78,]  0.6539641 13.518450 322.04642
##  [79,]  0.9253902 16.054590 313.59896
##  [80,]  0.7793590 16.841027 255.56883
##  [81,]  1.0194955 11.042410 295.35538
##  [82,]  0.7009293 13.792970 422.28623
##  [83,]  0.8424526 13.792970 448.40455
##  [84,]  0.8392194 13.792970 443.19267
##  [85,]  0.5411268 14.138606 418.34220
##  [86,]  0.4257133 11.813391 276.34270
##  [87,]  0.6528078 21.041419 752.47349
##  [88,]  0.4058400 10.015069 326.18105
##  [89,]  0.8477125 10.880199 491.14487
##  [90,]  0.4574645 13.220483 453.40358
##  [91,]  0.8266277 13.220483 428.78827
##  [92,]  0.4594026 14.172118 413.94161
##  [93,]  0.5878816 14.448077 281.10539
##  [94,]  1.0579385 14.448077 349.54302
##  [95,]  0.7206249 14.591671 633.72640
##  [96,]  0.8774119 14.591671 407.27045
##  [97,]  0.6309678 13.461079 515.62914
##  [98,]  0.9044731 14.150340 441.29861
##  [99,]  0.7180096  9.639470 409.25655
## [100,]  0.8183692  8.995749 472.88672
##
## , , 3
##
##               [,1]       [,2]      [,3]
##   [1,]  0.53529010   8.335203  134.4304
##   [2,]  0.69842287  11.133567  177.1993
##   [3,]  0.41531657   9.676353  232.9910
##   [4,]  0.40395334   5.847050  128.5498
##   [5,]  0.39508995   7.658995  170.0755
##   [6,]  1.02700736   9.797953  128.0982
##   [7,]  0.57271363   9.236724  171.3267
##   [8,]  0.55747195   9.842866  171.0200
##   [9,]  0.48769434   8.174354  127.8289
```

```
##  [10,]  0.83844788  11.811818  217.2316
##  [11,]  0.08819304   8.817826  140.4019
##  [12,]  1.02319769  14.687574  191.7547
##  [13,]  1.03940091  14.687574  184.5325
##  [14,]  0.05230825   8.185566  172.3612
##  [15,]  1.71221531  22.617271  294.2387
##  [16,]  1.35084262  22.617271  358.6287
##  [17,]  1.26321684  14.756395  172.0754
##  [18,]  1.28578801  23.147593  258.8735
##  [19,]  2.67785361  23.147593  150.5741
##  [20,]  3.77185760  33.924953  207.7252
##  [21,]  3.19690672  33.924953  223.3502
##  [22,]  4.51092882  33.924953  196.8159
##  [23,]  2.72380898  33.924953  249.9891
##  [24,]  1.90749012  19.292600  117.9325
##  [25,]  1.20499678  19.292600  161.5823
##  [26,]  1.55091379  23.925724  213.1750
##  [27,]  4.02051088  29.545917  145.7055
##  [28,]  2.54242207  29.545917  154.1955
##  [29,]  2.37600803  32.240888  299.9196
##  [30,]  4.34624561  39.356602  245.0118
##  [31,]  1.66698680  23.512595  266.6671
##  [32,]  4.52299472  27.692859  133.9001
##  [33,]  1.62983120  37.543127  525.0570
##  [34,]  2.40952848  45.223870  518.4067
##  [35,]  4.15298920  45.223870  226.9023
##  [36,]  2.09685922  35.867476  254.6936
##  [37,] 11.54652655  78.789303  399.9988
##  [38,]  2.79569015  29.955569  197.6801
##  [39,]  2.95013304  37.936964  237.3486
##  [40,]  6.03157529  43.159387  215.3606
##  [41,]  3.80375839  43.159387  285.2230
##  [42,] 10.16260298  57.815590  235.7281
##  [43,]  3.94006870  26.266917  117.4269
##  [44,]  2.31784859  27.635230  169.2536
##  [45,]  1.53974320  21.665111  164.5153
##  [46,]  3.44755537  38.139549  223.8765
##  [47,]  6.61233125  38.139549  149.9391
##  [48,]  2.29835245  23.357732  162.7095
##  [49,]  2.02358501  17.801162  113.4287
##  [50,]  1.70409412  23.619295  202.1886
##  [51,]  2.04741744  30.188380  246.6528
##  [52,]  2.68360631  24.277503  160.3883
##  [53,]  3.66957726  43.453279  351.8620
##  [54,]  7.19852922  50.602958  237.1548
##  [55,]  6.50076392  52.976757  244.5275
```

```
##  [56,]  8.60963812  52.076002  216.7829
##  [57,]  5.94170526  81.485669  475.0395
##  [58,] 12.39210130  81.975323  246.0523
##  [59,]  6.31691554  44.404987  167.9364
##  [60,]  8.54931656  39.389852   76.1895
##  [61,]  5.84510729  43.488153  152.1890
##  [62,] 18.66283619  63.709374  107.4656
##  [63,] 10.43983855 100.854824  470.0971
##  [64,]  9.94704323 100.854824  536.8995
##  [65,] 10.86051637  65.950131  247.2802
##  [66,]  7.99179413  65.950131  370.9823
##  [67,]  4.08792157  84.539805  942.9590
##  [68,]  7.76893866  83.057809  506.6569
##  [69,]  4.39184860  92.862695  884.6889
##  [70,]  8.40794064  92.862695  581.1110
##  [71,]  7.79588038  80.756697  369.0473
##  [72,] 14.48721720  80.756697  278.5689
##  [73,]  9.94440832  80.756697  425.3578
##  [74,]  8.58662316 143.433856  975.9504
##  [75,]  4.17039534  97.905276 1390.5400
##  [76,]  4.53594351 121.112846 2106.1161
##  [77,]  1.89901145  76.863297 1927.1307
##  [78,]  2.25987769  55.551272 1323.9276
##  [79,]  2.74105103  74.387976 1454.8837
##  [80,]  3.12713234  80.678801 1225.0209
##  [81,]  2.02504894  39.402213 1055.8239
##  [82,]  1.93410134  57.480760 1760.5597
##  [83,]  1.85161997  57.480760 1870.3780
##  [84,]  1.85765110  57.480760 1848.6812
##  [85,]  1.84236477  59.948482 1774.3175
##  [86,]  1.40210702  44.184107 1072.8432
##  [87,]  2.61013040 117.695071 4210.2914
##  [88,]  1.14565939  33.389067 1087.5199
##  [89,]  1.43241317  38.425252 1736.3310
##  [90,]  1.54085049  53.487772 1834.9273
##  [91,]  1.76208939  53.487772 1736.5990
##  [92,]  1.58984489  60.190007 1758.2954
##  [93,]  2.39238807  62.194024 1210.6415
##  [94,]  2.59820649  62.194024 1506.8214
##  [95,]  1.63037434  63.247435 2747.8328
##  [96,]  2.09735156  63.247435 1767.0197
##  [97,]  1.52153891  55.151460 2113.4527
##  [98,]  2.17764373  60.033005 1874.2031
##  [99,]  1.24995253  31.295395 1330.0151
## [100,]  1.18226042  27.840656 1465.0855
##
```

```
##
## attr(,"class")
## [1] "mixgpd_predict"
```

5.4. Conditional exceedance curves

Tail risk is often easier to interpret via exceedance probabilities $P(Y > y \mid x)$. We compute conditional survival curves over a grid of ozone values for the same profiles.

```
y_grid <- seq(min(y), max(y) * 1.3, length.out = 150)

pred_bulk_s <- predict(fit_bulk, newdata = X_prof, y = y_grid, type = "survival")
pred_gpd_s  <- predict(fit_gpd,  newdata = X_prof, y = y_grid, type = "survival")

pred_bulk_s

## $fit
##     id          y    survival        lower      upper
## 1    1   1.000000 0.332516624 1.208194e-02 0.93769759
## 2    1   2.459060 0.170122966 2.670186e-05 0.62993687
## 3    1   3.918121 0.121508364 1.628915e-07 0.49453245
## 4    1   5.377181 0.106956545 2.816264e-09 0.45825979
## 5    1   6.836242 0.095555394 2.503892e-11 0.45429107
## 6    1   8.295302 0.088703187 5.155495e-12 0.45095036
## 7    1   9.754362 0.082463897 4.416523e-13 0.39174851
## 8    1  11.213423 0.076317347 1.208200e-14 0.37748954
## 9    1  12.672483 0.071528826 2.425837e-15 0.36314369
## 10   1  14.131544 0.068351505 7.382983e-16 0.34903385
## 11   1  15.590604 0.065722578 2.109424e-16 0.33560593
## 12   1  17.049664 0.063342619 1.054712e-16 0.32707956
## 13   1  18.508725 0.061125322 0.000000e+00 0.31818929
## 14   1  19.967785 0.059039077 0.000000e+00 0.31029677
## 15   1  21.426846 0.057068900 0.000000e+00 0.30439040
## 16   1  22.885906 0.055205988 0.000000e+00 0.29805593
## 17   1  24.344966 0.053444129 0.000000e+00 0.29131459
## 18   1  25.804027 0.051778150 0.000000e+00 0.28420084
## 19   1  27.263087 0.050203222 0.000000e+00 0.27675821
## 20   1  28.722148 0.048714621 0.000000e+00 0.26903596
## 21   1  30.181208 0.047307681 0.000000e+00 0.26108621
## 22   1  31.640268 0.045977833 0.000000e+00 0.25296172
## 23   1  33.099329 0.044720659 0.000000e+00 0.24471422
## 24   1  34.558389 0.043531924 0.000000e+00 0.23639312
## 25   1  36.017450 0.042407596 0.000000e+00 0.22804463
## 26   1  37.476510 0.041343853 0.000000e+00 0.21971122
## 27   1  38.935570 0.040337073 0.000000e+00 0.21143125
## 28   1  40.394631 0.039383825 0.000000e+00 0.20323887
```

```
## 29   1  41.853691 0.038480860 0.000000e+00 0.19694819
## 30   1  43.312752 0.037625102 0.000000e+00 0.19134617
## 31   1  44.771812 0.036813641 0.000000e+00 0.18588621
## 32   1  46.230872 0.036043728 0.000000e+00 0.18057612
## 33   1  47.689933 0.035312773 0.000000e+00 0.17542171
## 34   1  49.148993 0.034618342 0.000000e+00 0.17042706
## 35   1  50.608054 0.033958153 0.000000e+00 0.16559470
## 36   1  52.067114 0.033330073 0.000000e+00 0.16092580
## 37   1  53.526174 0.032732112 0.000000e+00 0.15642041
## 38   1  54.985235 0.032162418 0.000000e+00 0.15207758
## 39   1  56.444295 0.031619274 0.000000e+00 0.14789552
## 40   1  57.903356 0.031101082 0.000000e+00 0.14387177
## 41   1  59.362416 0.030606364 0.000000e+00 0.14000328
## 42   1  60.821477 0.030133749 0.000000e+00 0.13638250
## 43   1  62.280537 0.029681966 0.000000e+00 0.13310066
## 44   1  63.739597 0.029249836 0.000000e+00 0.13076591
## 45   1  65.198658 0.028836263 0.000000e+00 0.13025179
## 46   1  66.657718 0.028440227 0.000000e+00 0.12974139
## 47   1  68.116779 0.028060779 0.000000e+00 0.12923464
## 48   1  69.575839 0.027697031 0.000000e+00 0.12873150
## 49   1  71.034899 0.027348154 0.000000e+00 0.12823192
## 50   1  72.493960 0.027013371 0.000000e+00 0.12773586
## 51   1  73.953020 0.026691952 0.000000e+00 0.12724327
## 52   1  75.412081 0.026383209 0.000000e+00 0.12675412
## 53   1  76.871141 0.026086496 0.000000e+00 0.12626838
## 54   1  78.330201 0.025801202 0.000000e+00 0.12578601
## 55   1  79.789262 0.025526751 0.000000e+00 0.12530697
## 56   1  81.248322 0.025262597 0.000000e+00 0.12483124
## 57   1  82.707383 0.025008224 0.000000e+00 0.12435879
## 58   1  84.166443 0.024763142 0.000000e+00 0.12388959
## 59   1  85.625503 0.024526889 0.000000e+00 0.12342360
## 60   1  87.084564 0.024299024 0.000000e+00 0.12296081
## 61   1  88.543624 0.024079132 0.000000e+00 0.12250119
## 62   1  90.002685 0.023866816 0.000000e+00 0.12204470
## 63   1  91.461745 0.023661701 0.000000e+00 0.12159134
## 64   1  92.920805 0.023463433 0.000000e+00 0.12114107
## 65   1  94.379866 0.023271673 0.000000e+00 0.12069386
## 66   1  95.838926 0.023086102 0.000000e+00 0.12024970
## 67   1  97.297987 0.022906417 0.000000e+00 0.11980857
## 68   1  98.757047 0.022732330 0.000000e+00 0.11937043
## 69   1 100.216107 0.022563569 0.000000e+00 0.11893526
## 70   1 101.675168 0.022399877 0.000000e+00 0.11850306
## 71   1 103.134228 0.022241008 0.000000e+00 0.11807378
## 72   1 104.593289 0.022086732 0.000000e+00 0.11764741
## 73   1 106.052349 0.021936829 0.000000e+00 0.11722393
## 74   1 107.511409 0.021791093 0.000000e+00 0.11680332
```

```
## 75   1 108.970470 0.021649327 0.000000e+00 0.11638555
## 76   1 110.429530 0.021511345 0.000000e+00 0.11597060
## 77   1 111.888591 0.021376973 0.000000e+00 0.11555845
## 78   1 113.347651 0.021246044 0.000000e+00 0.11514909
## 79   1 114.806711 0.021118401 0.000000e+00 0.11474248
## 80   1 116.265772 0.020993895 0.000000e+00 0.11433862
## 81   1 117.724832 0.020872386 0.000000e+00 0.11393747
## 82   1 119.183893 0.020753740 0.000000e+00 0.11353902
## 83   1 120.642953 0.020637833 0.000000e+00 0.11314324
## 84   1 122.102013 0.020524545 0.000000e+00 0.11275013
## 85   1 123.561074 0.020413763 0.000000e+00 0.11235964
## 86   1 125.020134 0.020305381 0.000000e+00 0.11197178
## 87   1 126.479195 0.020199299 0.000000e+00 0.11158651
## 88   1 127.938255 0.020095419 0.000000e+00 0.11120381
## 89   1 129.397315 0.019993653 0.000000e+00 0.11082367
## 90   1 130.856376 0.019893914 0.000000e+00 0.11044607
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## 92   1 133.774497 0.019700197 0.000000e+00 0.10969838
## 93   1 135.233557 0.019606069 0.000000e+00 0.10932827
## 94   1 136.692617 0.019513668 0.000000e+00 0.10896061
## 95   1 138.151678 0.019422928 0.000000e+00 0.10860717
## 96   1 139.610738 0.019333787 0.000000e+00 0.10825798
## 97   1 141.069799 0.019246186 0.000000e+00 0.10791119
## 98   1 142.528859 0.019160068 0.000000e+00 0.10756675
## 99   1 143.987919 0.019075380 0.000000e+00 0.10722466
## 100  1 145.446980 0.018992071 0.000000e+00 0.10688490
## 101  1 146.906040 0.018910093 0.000000e+00 0.10654744
## 102  1 148.365101 0.018829400 0.000000e+00 0.10621226
## 103  1 149.824161 0.018749947 0.000000e+00 0.10587934
## 104  1 151.283221 0.018671694 0.000000e+00 0.10554866
## 105  1 152.742282 0.018594600 0.000000e+00 0.10522021
## 106  1 154.201342 0.018518627 0.000000e+00 0.10489396
## 107  1 155.660403 0.018443739 0.000000e+00 0.10456988
## 108  1 157.119463 0.018369903 0.000000e+00 0.10424797
## 109  1 158.578523 0.018297084 0.000000e+00 0.10392821
## 110  1 160.037584 0.018225251 0.000000e+00 0.10361057
## 111  1 161.496644 0.018154374 0.000000e+00 0.10329503
## 112  1 162.955705 0.018084425 0.000000e+00 0.10298158
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## 117  1 170.251007 0.017747667 0.000000e+00 0.10144499
## 118  1 171.710067 0.017682744 0.000000e+00 0.10114369
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## 120  1 174.628188 0.017555149 0.000000e+00 0.10054693
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## 121  1 176.087248 0.017492438 0.000000e+00 0.10025146
## 122  1 177.546309 0.017430426 0.000000e+00 0.09995789
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## 129  1 187.759732 0.017014467 0.000000e+00 0.09795505
## 130  1 189.218792 0.016957448 0.000000e+00 0.09767618
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## 136  1 197.973154 0.016626627 0.000000e+00 0.09603954
## 137  1 199.432215 0.016573262 0.000000e+00 0.09577273
## 138  1 200.891275 0.016520375 0.000000e+00 0.09550757
## 139  1 202.350336 0.016467955 0.000000e+00 0.09524407
## 140  1 203.809396 0.016415993 0.000000e+00 0.09498220
## 141  1 205.268456 0.016364479 0.000000e+00 0.09472195
## 142  1 206.727517 0.016313404 0.000000e+00 0.09446331
## 143  1 208.186577 0.016262760 0.000000e+00 0.09420626
## 144  1 209.645638 0.016212537 0.000000e+00 0.09395079
## 145  1 211.104698 0.016162727 0.000000e+00 0.09369689
## 146  1 212.563758 0.016113323 0.000000e+00 0.09344455
## 147  1 214.022819 0.016064317 0.000000e+00 0.09319374
## 148  1 215.481879 0.016015701 0.000000e+00 0.09294446
## 149  1 216.940940 0.015967468 0.000000e+00 0.09269670
## 150  1 218.400000 0.015919611 0.000000e+00 0.09245044
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## 152  2   2.459060 0.062726434 3.170547e-03 0.15500826
## 153  2   3.918121 0.036277597 6.453261e-05 0.12641347
## 154  2   5.377181 0.026400335 2.206644e-06 0.10893260
## 155  2   6.836242 0.021189366 1.151592e-07 0.09578260
## 156  2   8.295302 0.017940813 8.407972e-09 0.08519522
## 157  2   9.754362 0.015702118 7.993950e-10 0.07698621
## 158  2  11.213423 0.014053154 9.393611e-11 0.07038608
## 159  2  12.672483 0.012780024 1.312764e-11 0.06494182
## 160  2  14.131544 0.011762223 2.119643e-12 0.06077878
## 161  2  15.590604 0.010926524 3.867240e-13 0.05771655
## 162  2  17.049664 0.010225738 7.842060e-14 0.05502459
## 163  2  18.508725 0.009628003 1.743050e-14 0.05266084
## 164  2  19.967785 0.009110974 4.135581e-15 0.05056852
## 165  2  21.426846 0.008658474 1.171285e-15 0.04867915
## 166  2  22.885906 0.008258485 2.636780e-16 0.04696208
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## 167  2  24.344966 0.007901870 1.054712e-16 0.04539281
## 168  2  25.804027 0.007581548 0.000000e+00 0.04395147
## 169  2  27.263087 0.007291940 0.000000e+00 0.04262173
## 170  2  28.722148 0.007028584 0.000000e+00 0.04139006
## 171  2  30.181208 0.006787864 0.000000e+00 0.04024511
## 172  2  31.640268 0.006566818 0.000000e+00 0.03917730
## 173  2  33.099329 0.006362991 0.000000e+00 0.03817846
## 174  2  34.558389 0.006174332 0.000000e+00 0.03724158
## 175  2  36.017450 0.005999111 0.000000e+00 0.03636062
## 176  2  37.476510 0.005835859 0.000000e+00 0.03553031
## 177  2  38.935570 0.005683318 0.000000e+00 0.03474608
## 178  2  40.394631 0.005540408 0.000000e+00 0.03409447
## 179  2  41.853691 0.005406190 0.000000e+00 0.03363018
## 180  2  43.312752 0.005279849 0.000000e+00 0.03318613
## 181  2  44.771812 0.005160669 0.000000e+00 0.03276082
## 182  2  46.230872 0.005048023 0.000000e+00 0.03235292
## 183  2  47.689933 0.004941356 0.000000e+00 0.03196122
## 184  2  49.148993 0.004840176 0.000000e+00 0.03158465
## 185  2  50.608054 0.004744044 0.000000e+00 0.03122220
## 186  2  52.067114 0.004652570 0.000000e+00 0.03087299
## 187  2  53.526174 0.004565402 0.000000e+00 0.03053620
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## 190  2  57.903356 0.004326732 0.000000e+00 0.02959318
## 191  2  59.362416 0.004253926 0.000000e+00 0.02929919
## 192  2  60.821477 0.004184123 0.000000e+00 0.02901444
## 193  2  62.280537 0.004117130 0.000000e+00 0.02873845
## 194  2  63.739597 0.004052769 0.000000e+00 0.02845916
## 195  2  65.198658 0.003990880 0.000000e+00 0.02815135
## 196  2  66.657718 0.003931313 0.000000e+00 0.02785347
## 197  2  68.116779 0.003873932 0.000000e+00 0.02756499
## 198  2  69.575839 0.003818612 0.000000e+00 0.02728543
## 199  2  71.034899 0.003765236 0.000000e+00 0.02701434
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## 203  2  76.871141 0.003569149 0.000000e+00 0.02600655
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## 205  2  79.789262 0.003480332 0.000000e+00 0.02554371
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## 207  2  82.707383 0.003396871 0.000000e+00 0.02510503
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## 348  3  69.575839 0.228419811 5.944964e-06 0.44880542
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## 350  3  72.493960 0.224420925 5.457343e-07 0.44871686
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## 354  3  78.330201 0.216716615 1.187553e-08 0.44854413
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## 356  3  81.248322 0.212983045 8.713519e-09 0.44844515
## 357  3  82.707383 0.211142468 7.694941e-09 0.44838453
## 358  3  84.166443 0.209318969 6.816808e-09 0.44831037
## 359  3  85.625503 0.207512440 6.050268e-09 0.44821575
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## 361  3  88.543624 0.203949749 4.788969e-09 0.44792470
## 362  3  90.002685 0.202193016 4.270310e-09 0.44770006
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## 364  3  92.920805 0.198726175 3.410104e-09 0.44699332
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## 366  3  95.838926 0.195316832 2.738177e-09 0.44576110
## 367  3  97.297987 0.193631991 2.044794e-09 0.44486365
## 368  3  98.757047 0.191959673 1.354397e-09 0.44372632
## 369  3 100.216107 0.190299604 8.981733e-10 0.44230664
## 370  3 101.675168 0.188651753 5.963571e-10 0.44064535
## 371  3 103.134228 0.187016344 3.964581e-10 0.43984624
## 372  3 104.593289 0.185393843 2.639030e-10 0.43984405
## 373  3 106.052349 0.183784930 1.758968e-10 0.43984139
## 374  3 107.511409 0.182190466 1.173943e-10 0.43983818
## 375  3 108.970470 0.180611441 7.845489e-11 0.43983433
## 376  3 110.429530 0.179048936 5.250296e-11 0.43982972
## 377  3 111.888591 0.177504078 3.518406e-11 0.43982422
## 378  3 113.347651 0.175978003 2.361086e-11 0.43981771
## 379  3 114.806711 0.174471827 1.586664e-11 0.43981001
## 380  3 116.265772 0.172986623 1.067750e-11 0.43980097
## 381  3 117.724832 0.171523402 7.195741e-12 0.43979039
## 382  3 119.183893 0.170083100 4.856232e-12 0.43977806
## 383  3 120.642953 0.168666567 3.282016e-12 0.43976375
## 384  3 122.102013 0.167274558 2.221345e-12 0.43974720
## 385  3 123.561074 0.165907729 1.505671e-12 0.43972814
## 386  3 125.020134 0.164566621 1.021921e-12 0.43970627
## 387  3 126.479195 0.163251662 6.946971e-13 0.43968127
## 388  3 127.938255 0.161963150 4.729439e-13 0.43965279
## 389  3 129.397315 0.160701255 3.224365e-13 0.43962046
## 390  3 130.856376 0.159466006 2.201295e-13 0.43958388
## 391  3 132.315436 0.158257294 1.504602e-13 0.43954262
## 392  3 133.774497 0.157074870 1.030509e-13 0.43949625
## 393  3 135.233557 0.155918349 7.067125e-14 0.43944427
## 394  3 136.692617 0.154787217 4.857503e-14 0.43938621
## 395  3 138.151678 0.153680843 3.338718e-14 0.43932152
## 396  3 139.610738 0.152598495 2.305101e-14 0.43924965
```

```
## 397  3 141.069799 0.151539356 1.587896e-14 0.43917005
## 398  3 142.528859 0.150502545 1.102729e-14 0.43908209
## 399  3 143.987919 0.149487142 7.652212e-15 0.43898518
## 400  3 145.446980 0.148492203 5.331846e-15 0.43887866
## 401  3 146.906040 0.147516792 3.697043e-15 0.43876189
## 402  3 148.365101 0.146559992 2.531308e-15 0.43863417
## 403  3 149.824161 0.145620931 1.793010e-15 0.43849481
## 404  3 151.283221 0.144698795 1.212919e-15 0.43834312
## 405  3 152.742282 0.143792839 8.437695e-16 0.43817835
## 406  3 154.201342 0.142902401 6.328271e-16 0.43799978
## 407  3 155.660403 0.142026902 4.218847e-16 0.43780668
## 408  3 157.119463 0.141165853 3.164136e-16 0.43759827
## 409  3 158.578523 0.140318848 2.109424e-16 0.43737382
## 410  3 160.037584 0.139485567 1.054712e-16 0.43713257
## 411  3 161.496644 0.138665761 1.054712e-16 0.43687376
## 412  3 162.955705 0.137859248 1.054712e-16 0.43659664
## 413  3 164.414765 0.137065902 5.273559e-17 0.43630045
## 414  3 165.873826 0.136285640 0.000000e+00 0.43598447
## 415  3 167.332886 0.135518409 0.000000e+00 0.43591180
## 416  3 168.791946 0.134764177 0.000000e+00 0.43589764
## 417  3 170.251007 0.134022923 0.000000e+00 0.43588174
## 418  3 171.710067 0.133294621 0.000000e+00 0.43586390
## 419  3 173.169128 0.132579235 0.000000e+00 0.43584395
## 420  3 174.628188 0.131876712 0.000000e+00 0.43582169
## 421  3 176.087248 0.131186972 0.000000e+00 0.43579690
## 422  3 177.546309 0.130509906 0.000000e+00 0.43576937
## 423  3 179.005369 0.129845375 0.000000e+00 0.43573885
## 424  3 180.464430 0.129193202 0.000000e+00 0.43570510
## 425  3 181.923490 0.128553178 0.000000e+00 0.43566784
## 426  3 183.382550 0.127925059 0.000000e+00 0.43562681
## 427  3 184.841611 0.127308572 0.000000e+00 0.43558170
## 428  3 186.300671 0.126703411 0.000000e+00 0.43553222
## 429  3 187.759732 0.126109247 0.000000e+00 0.43547804
## 430  3 189.218792 0.125525729 0.000000e+00 0.43541883
## 431  3 190.677852 0.124952490 0.000000e+00 0.43535424
## 432  3 192.136913 0.124389146 0.000000e+00 0.43528392
## 433  3 193.595973 0.123835308 0.000000e+00 0.43520748
## 434  3 195.055034 0.123290579 0.000000e+00 0.43512455
## 435  3 196.514094 0.122754562 0.000000e+00 0.43503473
## 436  3 197.973154 0.122226864 0.000000e+00 0.43493761
## 437  3 199.432215 0.121707095 0.000000e+00 0.43483276
## 438  3 200.891275 0.121194875 0.000000e+00 0.43471976
## 439  3 202.350336 0.120689835 0.000000e+00 0.43459817
## 440  3 203.809396 0.120191618 0.000000e+00 0.43446754
## 441  3 205.268456 0.119699882 0.000000e+00 0.43432741
## 442  3 206.727517 0.119214300 0.000000e+00 0.43402493
```

```
## 443  3 208.186577 0.118734562 0.000000e+00 0.43347029
## 444  3 209.645638 0.118260375 0.000000e+00 0.43289463
## 445  3 211.104698 0.117791462 0.000000e+00 0.43229743
## 446  3 212.563758 0.117327564 0.000000e+00 0.43167815
## 447  3 214.022819 0.116868439 0.000000e+00 0.43103628
## 448  3 215.481879 0.116413861 0.000000e+00 0.43037130
## 449  3 216.940940 0.115963619 0.000000e+00 0.42968269
## 450  3 218.400000 0.115517518 0.000000e+00 0.42896995
##
## $lower
## NULL
##
## $upper
## NULL
##
## $type
## [1] "survival"
##
## $grid
##   [1]   1.000000   2.459060   3.918121   5.377181   6.836242   8.295302
##   [7]   9.754362  11.213423  12.672483  14.131544  15.590604  17.049664
##  [13]  18.508725  19.967785  21.426846  22.885906  24.344966  25.804027
##  [19]  27.263087  28.722148  30.181208  31.640268  33.099329  34.558389
##  [25]  36.017450  37.476510  38.935570  40.394631  41.853691  43.312752
##  [31]  44.771812  46.230872  47.689933  49.148993  50.608054  52.067114
##  [37]  53.526174  54.985235  56.444295  57.903356  59.362416  60.821477
##  [43]  62.280537  63.739597  65.198658  66.657718  68.116779  69.575839
##  [49]  71.034899  72.493960  73.953020  75.412081  76.871141  78.330201
##  [55]  79.789262  81.248322  82.707383  84.166443  85.625503  87.084564
##  [61]  88.543624  90.002685  91.461745  92.920805  94.379866  95.838926
##  [67]  97.297987  98.757047 100.216107 101.675168 103.134228 104.593289
##  [73] 106.052349 107.511409 108.970470 110.429530 111.888591 113.347651
##  [79] 114.806711 116.265772 117.724832 119.183893 120.642953 122.102013
##  [85] 123.561074 125.020134 126.479195 127.938255 129.397315 130.856376
##  [91] 132.315436 133.774497 135.233557 136.692617 138.151678 139.610738
##  [97] 141.069799 142.528859 143.987919 145.446980 146.906040 148.365101
## [103] 149.824161 151.283221 152.742282 154.201342 155.660403 157.119463
## [109] 158.578523 160.037584 161.496644 162.955705 164.414765 165.873826
## [115] 167.332886 168.791946 170.251007 171.710067 173.169128 174.628188
## [121] 176.087248 177.546309 179.005369 180.464430 181.923490 183.382550
## [127] 184.841611 186.300671 187.759732 189.218792 190.677852 192.136913
## [133] 193.595973 195.055034 196.514094 197.973154 199.432215 200.891275
## [139] 202.350336 203.809396 205.268456 206.727517 208.186577 209.645638
## [145] 211.104698 212.563758 214.022819 215.481879 216.940940 218.400000
##
## attr(,"class")
```

```
## [1] "mixgpd_predict"

pred_gpd_s

## $fit
##     id          y     survival        lower        upper
## 1    1   1.000000 4.497894e-02 5.231334e-04 0.1097587484
## 2    1   2.459060 1.539848e-02 2.076147e-05 0.0480317953
## 3    1   3.918121 8.982597e-03 3.307567e-06 0.0308200878
## 4    1   5.377181 6.228602e-03 9.163449e-07 0.0225633526
## 5    1   6.836242 4.719039e-03 3.407780e-07 0.0177372053
## 6    1   8.295302 3.774117e-03 1.523289e-07 0.0145799598
## 7    1   9.754362 3.130654e-03 7.723896e-08 0.0123576584
## 8    1  11.213423 2.666120e-03 4.293101e-08 0.0107107662
## 9    1  12.672483 2.316041e-03 2.558967e-08 0.0094427006
## 10   1  14.131544 2.043385e-03 1.611564e-08 0.0084370332
## 11   1  15.590604 1.825420e-03 1.061014e-08 0.0076204765
## 12   1  17.049664 1.647452e-03 7.246035e-09 0.0069446289
## 13   1  18.508725 1.499575e-03 5.102928e-09 0.0063762557
## 14   1  19.967785 1.374879e-03 3.688824e-09 0.0058917908
## 15   1  21.426846 1.268400e-03 2.727282e-09 0.0054740652
## 16   1  22.885906 1.176486e-03 2.056257e-09 0.0051102788
## 17   1  24.344966 1.096390e-03 1.577225e-09 0.0047906967
## 18   1  25.804027 1.026010e-03 1.228345e-09 0.0045077848
## 19   1  27.263087 9.637097e-04 9.697059e-10 0.0042556232
## 20   1  28.722148 9.081976e-04 7.748998e-10 0.0040294961
## 21   1  30.181208 8.584407e-04 6.260645e-10 0.0038256017
## 22   1  31.640268 8.136038e-04 5.108760e-10 0.0036408409
## 23   1  33.099329 7.730044e-04 4.206773e-10 0.0034726630
## 24   1  34.558389 7.360794e-04 3.492880e-10 0.0033189496
## 25   1  36.017450 7.023604e-04 2.922296e-10 0.0031779269
## 26   1  37.476510 6.714544e-04 2.462130e-10 0.0030480986
## 27   1  38.935570 6.430298e-04 2.087916e-10 0.0029281937
## 28   1  40.394631 6.168045e-04 1.781251e-10 0.0028171260
## 29   1  41.853691 5.925373e-04 1.528143e-10 0.0027139614
## 30   1  43.312752 5.700206e-04 1.317844e-10 0.0026178925
## 31   1  44.771812 5.490750e-04 1.142030e-10 0.0025282179
## 32   1  46.230872 5.295444e-04 9.941887e-11 0.0024443251
## 33   1  47.689933 5.112927e-04 8.691949e-11 0.0023656773
## 34   1  49.148993 4.942004e-04 7.629748e-11 0.0022918017
## 35   1  50.608054 4.781624e-04 6.722758e-11 0.0022233536
## 36   1  52.067114 4.630858e-04 5.944776e-11 0.0021597257
## 37   1  53.526174 4.488881e-04 5.274610e-11 0.0020995943
## 38   1  54.985235 4.354961e-04 4.694966e-11 0.0020426801
## 39   1  56.444295 4.228442e-04 4.191696e-11 0.0019887327
## 40   1  57.903356 4.108738e-04 3.753151e-11 0.0019375270
```

```
## 41   1  59.362416 3.995322e-04 3.369683e-11 0.0018888600
## 42   1  60.821477 3.887721e-04 3.033279e-11 0.0018425483
## 43   1  62.280537 3.785505e-04 2.737230e-11 0.0017984255
## 44   1  63.739597 3.688288e-04 2.475921e-11 0.0017563409
## 45   1  65.198658 3.595718e-04 2.244621e-11 0.0017161568
## 46   1  66.657718 3.507476e-04 2.039333e-11 0.0016777480
## 47   1  68.116779 3.423269e-04 1.856639e-11 0.0016409998
## 48   1  69.575839 3.342833e-04 1.693670e-11 0.0016058073
## 49   1  71.034899 3.265923e-04 1.547941e-11 0.0015720742
## 50   1  72.493960 3.192317e-04 1.417317e-11 0.0015397118
## 51   1  73.953020 3.121811e-04 1.299994e-11 0.0015086387
## 52   1  75.412081 3.054215e-04 1.194386e-11 0.0014787796
## 53   1  76.871141 2.989356e-04 1.099133e-11 0.0014500650
## 54   1  78.330201 2.927075e-04 1.013049e-11 0.0014224307
## 55   1  79.789262 2.867222e-04 9.351084e-12 0.0013958171
## 56   1  81.248322 2.809662e-04 8.644060e-12 0.0013701689
## 57   1  82.707383 2.754267e-04 8.001738e-12 0.0013454347
## 58   1  84.166443 2.700920e-04 7.417036e-12 0.0013215667
## 59   1  85.625503 2.649510e-04 6.883974e-12 0.0012985204
## 60   1  87.084564 2.599936e-04 6.397288e-12 0.0012762542
## 61   1  88.543624 2.552103e-04 5.952197e-12 0.0012547294
## 62   1  90.002685 2.505923e-04 5.544590e-12 0.0012339098
## 63   1  91.461745 2.461312e-04 5.170847e-12 0.0012137612
## 64   1  92.920805 2.418194e-04 4.827508e-12 0.0011942521
## 65   1  94.379866 2.376496e-04 4.511777e-12 0.0011753525
## 66   1  95.838926 2.336149e-04 4.221071e-12 0.0011570343
## 67   1  97.297987 2.297092e-04 3.953085e-12 0.0011392714
## 68   1  98.757047 2.259263e-04 3.705686e-12 0.0011220390
## 69   1 100.216107 2.222606e-04 3.477055e-12 0.0011053137
## 70   1 101.675168 2.187070e-04 3.265496e-12 0.0010890735
## 71   1 103.134228 2.152604e-04 3.069631e-12 0.0010732978
## 72   1 104.593289 2.119161e-04 2.887932e-12 0.0010582487
## 73   1 106.052349 2.086697e-04 2.719294e-12 0.0010439019
## 74   1 107.511409 2.055171e-04 2.562620e-12 0.0010299423
## 75   1 108.970470 2.024543e-04 2.416925e-12 0.0010163543
## 76   1 110.429530 1.994776e-04 2.281217e-12 0.0010031233
## 77   1 111.888591 1.965835e-04 2.154793e-12 0.0009902354
## 78   1 113.347651 1.937686e-04 2.036871e-12 0.0009776774
## 79   1 114.806711 1.910299e-04 1.926800e-12 0.0009654368
## 80   1 116.265772 1.883642e-04 1.823974e-12 0.0009535016
## 81   1 117.724832 1.857688e-04 1.727737e-12 0.0009418606
## 82   1 119.183893 1.832409e-04 1.637698e-12 0.0009305029
## 83   1 120.642953 1.807780e-04 1.553319e-12 0.0009194183
## 84   1 122.102013 1.783777e-04 1.474318e-12 0.0009085972
## 85   1 123.561074 1.760377e-04 1.400147e-12 0.0008980301
## 86   1 125.020134 1.737557e-04 1.330477e-12 0.0008877082
```

```
## 87   1 126.479195 1.715296e-04 1.265088e-12 0.0008776230
## 88   1 127.938255 1.693575e-04 1.203546e-12 0.0008677666
## 89   1 129.397315 1.672374e-04 1.145734e-12 0.0008581312
## 90   1 130.856376 1.651676e-04 1.091160e-12 0.0008487093
## 91   1 132.315436 1.631462e-04 1.039879e-12 0.0008394941
## 92   1 133.774497 1.611717e-04 9.915097e-13 0.0008304786
## 93   1 135.233557 1.592425e-04 9.457768e-13 0.0008216566
## 94   1 136.692617 1.573570e-04 9.026752e-13 0.0008130218
## 95   1 138.151678 1.555137e-04 8.619938e-13 0.0008045683
## 96   1 139.610738 1.537114e-04 8.235579e-13 0.0007962905
## 97   1 141.069799 1.519487e-04 7.871565e-13 0.0007881829
## 98   1 142.528859 1.502243e-04 7.527312e-13 0.0007802403
## 99   1 143.987919 1.485369e-04 7.201184e-13 0.0007724577
## 100  1 145.446980 1.468855e-04 6.892598e-13 0.0007648303
## 101  1 146.906040 1.452690e-04 6.599388e-13 0.0007573535
## 102  1 148.365101 1.436862e-04 6.322665e-13 0.0007500229
## 103  1 149.824161 1.421360e-04 6.059098e-13 0.0007428343
## 104  1 151.283221 1.406177e-04 5.808742e-13 0.0007357834
## 105  1 152.742282 1.391301e-04 5.571543e-13 0.0007289047
## 106  1 154.201342 1.376723e-04 5.345335e-13 0.0007221668
## 107  1 155.660403 1.362436e-04 5.131173e-13 0.0007155539
## 108  1 157.119463 1.348430e-04 4.926948e-13 0.0007090627
## 109  1 158.578523 1.334697e-04 4.733075e-13 0.0007026898
## 110  1 160.037584 1.321230e-04 4.548611e-13 0.0006964320
## 111  1 161.496644 1.308021e-04 4.372336e-13 0.0006902862
## 112  1 162.955705 1.295063e-04 4.204304e-13 0.0006842494
## 113  1 164.414765 1.282349e-04 4.044515e-13 0.0006783187
## 114  1 165.873826 1.269872e-04 3.892442e-13 0.0006724913
## 115  1 167.332886 1.257626e-04 3.746919e-13 0.0006667646
## 116  1 168.791946 1.245604e-04 3.608003e-13 0.0006611360
## 117  1 170.251007 1.233801e-04 3.475109e-13 0.0006556029
## 118  1 171.710067 1.222210e-04 3.348821e-13 0.0006501629
## 119  1 173.169128 1.210826e-04 3.228029e-13 0.0006448138
## 120  1 174.628188 1.199644e-04 3.112149e-13 0.0006395531
## 121  1 176.087248 1.188659e-04 3.001766e-13 0.0006343788
## 122  1 177.546309 1.177864e-04 2.895822e-13 0.0006292887
## 123  1 179.005369 1.167256e-04 2.794209e-13 0.0006242808
## 124  1 180.464430 1.156830e-04 2.697564e-13 0.0006193531
## 125  1 181.923490 1.146581e-04 2.604778e-13 0.0006145036
## 126  1 183.382550 1.136505e-04 2.515793e-13 0.0006097306
## 127  1 184.841611 1.126598e-04 2.430667e-13 0.0006050322
## 128  1 186.300671 1.116855e-04 2.348871e-13 0.0006004067
## 129  1 187.759732 1.107272e-04 2.270351e-13 0.0005958523
## 130  1 189.218792 1.097846e-04 2.195105e-13 0.0005913675
## 131  1 190.677852 1.088572e-04 2.123718e-13 0.0005869507
## 132  1 192.136913 1.079448e-04 2.053968e-13 0.0005826003
```

```
## 133  1 193.595973 1.070469e-04 1.987549e-13 0.0005783148
## 134  1 195.055034 1.061633e-04 1.923295e-13 0.0005740928
## 135  1 196.514094 1.052935e-04 1.861789e-13 0.0005699329
## 136  1 197.973154 1.044374e-04 1.803557e-13 0.0005658337
## 137  1 199.432215 1.035945e-04 1.746492e-13 0.0005617939
## 138  1 200.891275 1.027645e-04 1.692119e-13 0.0005578122
## 139  1 202.350336 1.019473e-04 1.639383e-13 0.0005538874
## 140  1 203.809396 1.011424e-04 1.588868e-13 0.0005500182
## 141  1 205.268456 1.003496e-04 1.540518e-13 0.0005462034
## 142  1 206.727517 9.956867e-05 1.494388e-13 0.0005424420
## 143  1 208.186577 9.879933e-05 1.449368e-13 0.0005387328
## 144  1 209.645638 9.804132e-05 1.405986e-13 0.0005350746
## 145  1 211.104698 9.729440e-05 1.363715e-13 0.0005314666
## 146  1 212.563758 9.655834e-05 1.323081e-13 0.0005279076
## 147  1 214.022819 9.583289e-05 1.284611e-13 0.0005243966
## 148  1 215.481879 9.511784e-05 1.246725e-13 0.0005209327
## 149  1 216.940940 9.441296e-05 1.211059e-13 0.0005175149
## 150  1 218.400000 9.371805e-05 1.176448e-13 0.0005141424
## 151  2   1.000000 4.679283e-01 4.064627e-01 0.5025940949
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## 398  3 142.528859 5.361095e-02 6.952143e-03 0.2004427321
## 399  3 143.987919 5.303389e-02 6.762352e-03 0.1984644414
## 400  3 145.446980 5.246828e-02 6.617039e-03 0.1965187148
## 401  3 146.906040 5.191379e-02 6.461028e-03 0.1946048047
## 402  3 148.365101 5.137012e-02 6.310047e-03 0.1927219851
## 403  3 149.824161 5.083695e-02 6.163896e-03 0.1908695515
## 404  3 151.283221 5.031399e-02 6.022382e-03 0.1890468198
## 405  3 152.742282 4.980096e-02 5.885323e-03 0.1872531254
## 406  3 154.201342 4.929758e-02 5.752547e-03 0.1854878231
## 407  3 155.660403 4.880359e-02 5.622200e-03 0.1837502857
## 408  3 157.119463 4.831874e-02 5.494464e-03 0.1820399039
```

```
## 409  3 158.578523 4.784278e-02 5.370680e-03 0.1803560856
## 410  3 160.037584 4.737548e-02 5.250698e-03 0.1786982550
## 411  3 161.496644 4.691661e-02 5.134372e-03 0.1770658524
## 412  3 162.955705 4.646594e-02 5.021566e-03 0.1754583337
## 413  3 164.414765 4.602326e-02 4.912148e-03 0.1738751695
## 414  3 165.873826 4.558838e-02 4.805991e-03 0.1723158447
## 415  3 167.332886 4.516108e-02 4.702975e-03 0.1707798585
## 416  3 168.791946 4.474118e-02 4.602986e-03 0.1692667233
## 417  3 170.251007 4.432849e-02 4.502822e-03 0.1677759647
## 418  3 171.710067 4.392283e-02 4.392791e-03 0.1663071208
## 419  3 173.169128 4.352403e-02 4.286138e-03 0.1648597420
## 420  3 174.628188 4.313192e-02 4.182738e-03 0.1634333904
## 421  3 176.087248 4.274634e-02 4.082474e-03 0.1620276396
## 422  3 177.546309 4.236713e-02 3.985230e-03 0.1606420743
## 423  3 179.005369 4.199413e-02 3.890897e-03 0.1592762899
## 424  3 180.464430 4.162720e-02 3.799372e-03 0.1579298922
## 425  3 181.923490 4.126619e-02 3.710554e-03 0.1566024970
## 426  3 183.382550 4.091097e-02 3.624346e-03 0.1552937300
## 427  3 184.841611 4.056140e-02 3.540657e-03 0.1540032261
## 428  3 186.300671 4.021735e-02 3.459398e-03 0.1527306296
## 429  3 187.759732 3.987870e-02 3.380485e-03 0.1514755936
## 430  3 189.218792 3.954531e-02 3.303837e-03 0.1502377797
## 431  3 190.677852 3.921708e-02 3.251223e-03 0.1490168580
## 432  3 192.136913 3.889388e-02 3.200776e-03 0.1478125065
## 433  3 193.595973 3.857560e-02 3.151527e-03 0.1466244114
## 434  3 195.055034 3.826214e-02 3.103437e-03 0.1454522662
## 435  3 196.514094 3.795339e-02 3.056472e-03 0.1442957718
## 436  3 197.973154 3.764925e-02 3.010598e-03 0.1431546366
## 437  3 199.432215 3.734961e-02 2.965782e-03 0.1420285757
## 438  3 200.891275 3.705438e-02 2.921991e-03 0.1409173109
## 439  3 202.350336 3.676346e-02 2.879197e-03 0.1398205708
## 440  3 203.809396 3.647677e-02 2.836019e-03 0.1387380903
## 441  3 205.268456 3.619420e-02 2.791652e-03 0.1376696104
## 442  3 206.727517 3.591569e-02 2.748344e-03 0.1366148782
## 443  3 208.186577 3.564114e-02 2.706063e-03 0.1355736467
## 444  3 209.645638 3.537047e-02 2.664779e-03 0.1345456746
## 445  3 211.104698 3.510360e-02 2.623231e-03 0.1335307259
## 446  3 212.563758 3.484046e-02 2.575241e-03 0.1325285704
## 447  3 214.022819 3.458096e-02 2.523002e-03 0.1315389826
## 448  3 215.481879 3.432503e-02 2.472097e-03 0.1305617425
## 449  3 216.940940 3.407261e-02 2.422485e-03 0.1295966348
## 450  3 218.400000 3.382362e-02 2.374127e-03 0.1286434493
##
## $lower
## NULL
##
```

```
## $upper
## NULL
##
## $type
## [1] "survival"
##
## $grid
##   [1]   1.000000   2.459060   3.918121   5.377181   6.836242   8.295302
##   [7]   9.754362  11.213423  12.672483  14.131544  15.590604  17.049664
##  [13]  18.508725  19.967785  21.426846  22.885906  24.344966  25.804027
##  [19]  27.263087  28.722148  30.181208  31.640268  33.099329  34.558389
##  [25]  36.017450  37.476510  38.935570  40.394631  41.853691  43.312752
##  [31]  44.771812  46.230872  47.689933  49.148993  50.608054  52.067114
##  [37]  53.526174  54.985235  56.444295  57.903356  59.362416  60.821477
##  [43]  62.280537  63.739597  65.198658  66.657718  68.116779  69.575839
##  [49]  71.034899  72.493960  73.953020  75.412081  76.871141  78.330201
##  [55]  79.789262  81.248322  82.707383  84.166443  85.625503  87.084564
##  [61]  88.543624  90.002685  91.461745  92.920805  94.379866  95.838926
##  [67]  97.297987  98.757047 100.216107 101.675168 103.134228 104.593289
##  [73] 106.052349 107.511409 108.970470 110.429530 111.888591 113.347651
##  [79] 114.806711 116.265772 117.724832 119.183893 120.642953 122.102013
##  [85] 123.561074 125.020134 126.479195 127.938255 129.397315 130.856376
##  [91] 132.315436 133.774497 135.233557 136.692617 138.151678 139.610738
##  [97] 141.069799 142.528859 143.987919 145.446980 146.906040 148.365101
## [103] 149.824161 151.283221 152.742282 154.201342 155.660403 157.119463
## [109] 158.578523 160.037584 161.496644 162.955705 164.414765 165.873826
## [115] 167.332886 168.791946 170.251007 171.710067 173.169128 174.628188
## [121] 176.087248 177.546309 179.005369 180.464430 181.923490 183.382550
## [127] 184.841611 186.300671 187.759732 189.218792 190.677852 192.136913
## [133] 193.595973 195.055034 196.514094 197.973154 199.432215 200.891275
## [139] 202.350336 203.809396 205.268456 206.727517 208.186577 209.645638
## [145] 211.104698 212.563758 214.022819 215.481879 216.940940 218.400000
##
## attr(,"class")
## [1] "mixgpd_predict"
```

5.5. Interpretation

Across profiles, the bulk-only mixture typically captures central quantiles well but may be sensitive in high-quantile extrapolation depending on the kernel behavior in the far tail. The bulk+GPD specification enforces an extreme-value tail form beyond the learned threshold, which can stabilize inference for the 0.95–0.99 quantiles and exceedance probabilities. In applications where decisions depend on rare high outcomes (e.g., risk, reliability, safety), this tail regularization is often the main practical value of the model (Pickands 1975; de Haan and Ferreira 2006).

6. Benchmark application: Heavy-tailed positive data

In addition to conditional modeling, **DPmixGPD** is effective for standard density estimation tasks where the data generating process is complex. We analyze the `nc_pos200_k3` dataset included in the package, which consists of 200 positive-valued observations from a three-component mixture with a heavy tail.

6.1. Model specification

Since the data are strictly positive, we employ a Gamma kernel for the Dirichlet Process mixture bulk. We enable the GPD tail option to robustly capture the extreme observations.

```
# Load package data
data("nc_pos200_k3", package = "DPmixGPD")
y_bench <- nc_pos200_k3

# Configure the model bundle
# - 'sb' backend for stick-breaking approximation
# - 'gamma' kernel appropriate for positive data
# - 'GPD = TRUE' to enable tail splicing
bundle_bench <- build_nimble_bundle(
  y = y_bench,
  backend = "sb",
  kernel = "gamma",
  GPD = TRUE,
  components =6,
  mcmc = list(niter = 300, nburnin = 100, thin = 1, nchains = 1, seed = 123)
)

## Error in `build_nimble_bundle()`:
## !  'list' object cannot be coerced to type 'double'

# Run MCMC
fit_bench <- run_mcmc_bundle_manual(bundle_bench, show_progress = FALSE)

## Error:
## !  object 'bundle_bench' not found
```

6.2. Diagnostics and results

The summary output provides the posterior mean and credible intervals for the GPD shape parameter ξ. A positive ξ confirms the detection of heavy-tailed behavior in the data.

```
# Print summary to see tail index (xi) and threshold (u)
print(fit_bench)

## Error:
## !  object 'fit_bench' not found
```

```
summary(fit_bench)

## Error:
## !  object 'fit_bench' not found
```

We can visualize the fit by comparing the posterior predictive density against the histogram of the observed data. The combination of the Gamma mixture for the bulk and the GPD for the tail provides a smooth fit that adheres closely to the data across the entire support.

```
# Plot posterior predictive density
plot(fit_bench, type = "density",
     main = "Posterior Predictive Fit: Gamma-DPM + GPD")

## Error:
## !  object 'fit_bench' not found
```

7. Discussion and future work

DPmixGPD provides a unified Bayesian workflow for flexible bulk modeling and principled tail inference. Compared to purely parametric bulk+tail approaches, the DPM bulk reduces risk of misspecifying central features such as multimodality or skewness. Compared to bulk-only nonparametric mixtures, the explicit GPD tail can reduce instability in extreme quantiles. Planned extensions include: (i) additional kernels and regression-link options, (ii) improved posterior predictive diagnostics for tail adequacy, (iii) computational refinements for larger datasets (e.g., faster sampling or approximate inference), and (iv) expanded causal and survival modeling modules where tail behavior is scientifically central.

Computational details

All examples were run in R with **DPmixGPD**. MCMC settings in the manuscript are chosen for illustrative runtime; for substantive analysis, longer runs and convergence checks are recommended.

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Neal RM (2000). "Markov Chain Sampling Methods for Dirichlet Process Mixture Models." *Journal of Computational and Graphical Statistics*, **9**(2), 249–265.

Pickands J (1975). *Statistical Inference Using Extreme Order Statistics*. University of Copenhagen.

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